

COMSATS University Islamabad, Pakistan

Pulse Point

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Executive Summary

Pulse Point is a centralized, AI-assisted healthcare management system that brings patients, doctors, and administrators onto a single digital platform. The system addresses the common problem of fragmented healthcare processes where appointments, medical records, prescriptions, ambulance requests, pharmacy orders, laboratory tests, and patient history are stored separately or handled manually. By centralizing these functions, Pulse Point ensures secure storage, easy accessibility, and efficient management of all healthcare information.

Patients can create accounts, search for doctors by specialty, location, or name, and book appointments digitally. They can also request emergency ambulances with real-time GPS tracking, order medicines from the pharmacy with prescription upload capability, book laboratory tests and receive digital results, consult the AI-powered medical chatbot for instant guidance, and store medical records, lab reports, prescriptions, and consultation history securely in the cloud. Doctors can access these records only after proper authentication, ensuring privacy and patient consent.

Doctors use a centralized dashboard to view appointments, patient history, consultation details, and generate digital prescriptions. The prescription module is linked with the admin panel, which manages standardized medicine and dosage data. Doctors can also add clinical notes and update patient histories within the same system, creating a complete digital record over time.

The admin panel supports system integrity by handling doctor approvals, user verification, medicine inventory management, analytics, and overall monitoring of system activity. All communication between the client and server is secured using HTTPS protocols and authentication tokens to protect data integrity and user privacy.

Overall, Pulse Point modernizes healthcare by integrating essential services into a single cloud-based ecosystem, reducing paperwork, improving coordination, supporting informed medical decision-making, and providing real-time emergency response capabilities. The system is built to be scalable, reliable, and user-friendly, making it suitable for deployment in clinics, hospitals, and healthcare networks across Pakistan.

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Abbreviations

API	Application Programming Interface
UI	User Interface
UX	User Experience
EMR	Electronic Medical Records
EHR	Electronic Health Records
HMS	Hospital Management System
AI	Artificial Intelligence
GPS	Global Positioning System
OTP	One-Time Password
DB	Database
SDLC	Software Development Life Cycle
MVC	Model-View-Controller
NFR	Non-Functional Requirements
FR	Functional Requirements

CRUD	Create, Read, Update, Delete
JSON	JavaScript Object Notation
HTTPS	Hypertext Transfer Protocol Secure
FCM	Firebase Cloud Messaging
REST	Representational State Transfer

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Chapter 1

INTRODUCTION

1. Introduction

Healthcare in Pakistan faces long-standing challenges related to accessibility, coordination, and the secure handling of patient information. A significant portion of healthcare facilities, especially in small cities and rural areas, continue to rely on manual record-keeping methods, handwritten prescriptions, and paper-based appointment systems. As a result, patient histories, laboratory reports, and medical records are often scattered, misplaced, or unavailable when required. These gaps create delays in diagnosis, increase the chances of medical errors, and make it difficult for patients and healthcare providers to maintain continuity of care. With the rapid growth of population, overburdened public hospitals, and limited digital infrastructure in many medical institutions, the need for modern, centralized healthcare solutions in Pakistan has become increasingly urgent.

In recent years, Pakistan has seen a growing shift toward digital services across banking, e-commerce, and transportation. However, the healthcare sector still lacks a unified digital ecosystem that brings patients, doctors, and administrators together. While some private hospitals have introduced electronic systems, they remain isolated within individual institutions and do not offer a nationwide or publicly accessible platform. The absence of centralized digital health records means that when a patient visits different clinics or specialists, their medical data does not follow them. This leads to repeated tests, inconsistent diagnoses, and increased costs for patients. A cloud-based healthcare system has the potential to address these issues by storing medical information centrally and making it accessible through secure digital platforms, regardless of location or device.

Pulse Point is developed as a modern solution to these challenges within the context of Pakistan's healthcare environment. It is a comprehensive AI-assisted healthcare management system designed to integrate all major healthcare activities into a single cloud-based platform. The purpose of the system is to digitalize workflows, reduce administrative burdens, improve communication, and make healthcare more accessible to people across Pakistan. Whether a patient is in a large city like Karachi, Lahore, or Islamabad, or in a smaller district, Pulse Point enables them to access

healthcare services, maintain medical histories, and communicate with doctors without being limited by physical distance or fragmented record-keeping practices.

The system consists of a cross-platform mobile application for patients and doctors, and an admin panel that oversees operations. It is built using modern technologies such as Flutter for the frontend and Firebase for backend services, ensuring fast responsiveness, reliability, and scalability. Firebase Firestore is used as the database to store healthcare data, offering flexibility for handling patient profiles, reports, prescriptions, ambulance requests, pharmacy orders, and other structured and unstructured medical information. These technologies support the system's ability to serve a large and diverse user base, which is essential in a country where millions of people require accessible and efficient healthcare solutions.

Pulse Point enables patients in Pakistan to create digital accounts, search for doctors based on specialization, city, consultation fees, and clinic availability, and book appointments online. The system also allows patients to request emergency ambulances with real-time GPS tracking, providing the nearest available ambulance and real-time location updates until arrival. This feature is critical in a country where emergency response times are often delayed due to lack of coordination and real-time information.

The platform also empowers patients by enabling them to store their medical reports, lab results, prescriptions, and past consultation notes digitally. Through secure role-based authentication, patients can allow doctors to access their medical records securely, ensuring privacy and consent-driven data sharing. The integrated AI-powered medical assistant, powered by the Gemini API, provides intelligent symptom checks, basic medical advice, and instant responses to common health-related queries. By guiding patients toward appropriate services such as consulting a doctor, booking a lab test, or ordering a needed medicine, the AI assistant reduces unnecessary hospital visits and streamlines patient decision-making.

Doctors in Pakistan benefit from a dedicated dashboard where they can manage appointments, review patient histories, and generate prescriptions in a standardized format. Many clinics in Pakistan still rely on handwritten prescriptions, which can sometimes be unclear or inconsistent. Pulse Point solves this by providing a structured prescription generation module that is linked to an admin-controlled medicine database. Doctors can also add clinical notes and update patient histories, creating a more complete digital record over time.

The admin panel plays a central role in the ecosystem by verifying doctor credentials, monitoring system activity, managing medicine databases, and ensuring that all information remains authentic and secure. This is particularly important in Pakistan, where the verification of medical practitioners and the prevention of fraudulent activities are significant concerns.

Overall, Pulse Point aims to modernize Pakistan's healthcare system by introducing a cloud-based digital environment where patients, doctors, and administrators can interact seamlessly. The system reduces dependence on manual methods, minimizes the risks of lost or incomplete medical records, and provides a platform that supports remote access and secure communication. By addressing the unique challenges faced in Pakistan's healthcare sector, Pulse Point represents a practical and scalable solution that has the potential to enhance healthcare accessibility, transparency, and quality across the country.

1.1 Brief

The Pulse Point project is designed as a centralized healthcare management system that connects patients, doctors, and administrators through a unified digital ecosystem. The aim of the project is to overcome the limitations of traditional healthcare systems, where medical data, prescriptions, appointment details, ambulance requests, and other critical information are scattered across different platforms or handled manually. Pulse Point consolidates all essential healthcare services into one platform, ensuring secure storage, easy accessibility, accurate communication, and efficient workflows for all users. The system leverages modern mobile and cloud technologies to provide a reliable, scalable, and user-friendly healthcare experience.

Unlike traditional hospital management systems that are often expensive, complex, and designed for desktop computers, Pulse Point is built as a mobile-first solution that can be accessed from anywhere using a smartphone. This makes it particularly suitable for the Pakistani context, where smartphone penetration is high but desktop computer access is limited. The system is designed to be intuitive and easy to use, requiring minimal training for both patients and healthcare providers.

1.1.1 Purpose of the Project

The primary purpose of Pulse Point is to centralize healthcare operations by offering an all-in-one solution for appointment booking, emergency ambulance dispatch, medical record management,

digital prescription handling, pharmacy ordering, laboratory test booking, and AI-powered medical assistance. The project aims to simplify interactions between patients and healthcare providers by removing communication gaps and reducing the need for physical paperwork.

The SRS document serves as a formal contract among stakeholders, ensuring that all expectations and functionalities are clearly defined. It provides a structured foundation for design, implementation, verification, and maintenance phases of the project. The purpose is not only to develop a functional software system but also to demonstrate the application of software engineering principles, database design concepts, mobile development techniques, and AI integration skills learned throughout the BCS program.

Furthermore, the purpose extends beyond academic requirements. Pulse Point aims to make a positive social impact by improving healthcare accessibility, reducing emergency response times, preventing loss of medical records, and empowering patients to take control of their health information. By providing a platform that integrates all essential healthcare services, Pulse Point seeks to transform the way healthcare is delivered in Pakistan.

1.1.2 Scope of the System

The scope of Pulse Point extends across multiple platforms, including a cross-platform mobile application for patients and doctors, and an admin panel for system-wide data management. The project handles various core functionalities such as doctor discovery based on specialization and location, appointment scheduling, clinical history management, prescription generation, report viewing, real-time ambulance dispatch with GPS tracking, pharmacy ordering with inventory management, laboratory test booking with digital result delivery, AI-powered medical chatbot, and secure role-based access to sensitive patient documents.

The system is designed to serve three primary user categories:

Patients can create accounts, search for doctors, book appointments, request emergency ambulances with real-time tracking, order medicines from the pharmacy, upload prescription images for prescription-only medicines, book laboratory tests, consult the AI chatbot for medical guidance, view their complete medical history including past appointments, prescriptions, and lab results, and manage their profile information.

Doctors can log in to view their daily and upcoming appointments, access patient medical histories before consultations, write digital prescriptions using a standardized format linked to the admin-managed medicine database, add consultation notes to patient records, and update their profile information including availability and consultation fees.

Administrators can view all registered users, verify doctor registrations by reviewing uploaded credentials, approve or reject doctor applications, manage the medicine database (add, update, delete medicines, manage stock levels), view system logs to monitor user activity and detect suspicious behavior, and generate reports on system usage and performance.

By covering end-to-end healthcare processes, the system aims to serve as a complete digital solution for clinics, doctors, and patients. The system operates as a standalone product but is extensible to integrate with third-party health services through secure APIs.

1.1.3 Technologies Used

The Pulse Point system is developed using Flutter and Dart for cross-platform mobile functionality, which provides a modular and scalable structure suitable for large applications. Flutter allows the development of a single codebase that compiles to both Android and iOS platforms, significantly reducing development time and maintenance effort.

Firebase serves as the backend platform, providing multiple integrated services:

- **Firebase Authentication** handles user registration, login, password reset, email verification, and session management. It uses secure hashing algorithms (bcrypt) to store passwords and implements rate limiting to prevent brute-force attacks.
- **Cloud Firestore** is a flexible, scalable, NoSQL cloud database used to store all application data including user profiles, patient records, doctor information, appointments, ambulance requests, pharmacy orders, medicines inventory, lab tests, and medical records. Firestore provides real-time listeners that automatically push updates to connected clients, eliminating the need for manual refresh.
- **Firebase Cloud Messaging (FCM)** is a cross-platform messaging solution that enables the backend to send push notifications to mobile devices. Notifications are sent for

appointment confirmations, order status updates, ambulance dispatch alerts, and lab result availability.

- **Firebase Cloud Storage** is used to store binary files such as prescription images uploaded by patients, lab result PDFs uploaded by lab staff, and user profile pictures. Files are stored securely with access controlled by security rules.

The system also integrates two external APIs:

- **Gemini API by Google AI** provides access to Google's advanced large language model. The API is used to power the AI medical assistant chatbot, which responds to patient queries with helpful medical information while respecting safety guidelines and providing appropriate disclaimers.
- **Google Maps API** provides mapping, geocoding (converting addresses to GPS coordinates), distance calculation, and real-time location tracking. For the ambulance dispatch feature, the patient's GPS coordinates are captured, the nearest available ambulance is calculated using distance algorithms, and real-time tracking is provided on an interactive map.

Together, these technologies form a strong foundation for building a cloud-based healthcare platform that is secure, scalable, real-time, and cross-platform.

1.1.4 System Overview

Pulse Point operates as an interconnected system that supports users at various levels. The system architecture follows a client-server model with clear separation of concerns.

For Patients: After downloading the app and creating an account, patients log in to access their personalized dashboard. The dashboard displays quick-access cards for each major feature: Book Appointment, Request Ambulance, Pharmacy, Lab Tests, AI ChatBot, and Medical Records. A profile icon provides access to account settings and help documentation. Recent activity such as upcoming appointments and pending orders is displayed at the bottom of the dashboard for easy reference.

When booking an appointment, patients can search for doctors by name, specialization, or clinic location. The system displays doctor profiles including experience, consultation fee, and available time slots. Available slots are shown in 30-minute increments between 9 AM and 5 PM, Monday through Saturday. The system checks availability in real-time and prevents double-booking by using atomic Firestore transactions.

For emergency ambulance requests, patients can trigger the feature with a single tap, even without logging in. The system requests GPS permission, captures the patient's current location, displays it on a map with an option to adjust, allows selection of ambulance type (Basic/Advanced/ICU), dispatches the nearest available ambulance by calculating distances using the Haversine formula, and provides a real-time tracking screen showing the ambulance's location on a map with estimated arrival time.

The pharmacy module allows patients to browse medicines by category (Pain Relief, Antibiotics, Vitamins, Chronic Disease, etc.), view detailed information including price, stock status, and prescription requirement, add medicines to a shopping cart, upload prescription images for prescription-only medicines, place orders, and track order status through stages: Pending → Confirmed → Processing → Out for Delivery → Delivered.

The laboratory module allows patients to view available tests with descriptions and fees, select a test, choose an available date and time slot, confirm booking, and later view digital results when uploaded by lab staff. Notifications are sent when results are available.

The AI chatbot provides instant medical guidance for health-related queries. Patients type questions in natural language, and the system sends the query to the Gemini API, which returns a helpful response. A prominent disclaimer reminds users that the AI is not a substitute for professional medical advice and that emergencies require immediate contact with emergency services.

The medical records module provides a centralized repository of all patient health information, organized into tabs for Past Appointments, Prescriptions, and Lab Results. Each record can be viewed in detail, and files can be downloaded for sharing with other providers.

For Doctors: After logging in, doctors see a dashboard showing today's appointments list, including patient names, scheduled times, and status. Clicking on an appointment opens the

patient's medical history, including past consultations, prescriptions, and lab results. This information helps doctors make informed decisions during consultations. After seeing the patient, doctors can write a digital prescription using a structured form. They can select medicines from the admin-managed database, specify dosage, duration, and special instructions. The prescription is automatically saved to the patient's medical record and can be printed or shared electronically.

For Administrators: The admin panel provides oversight of the entire system. Administrators can view and manage all registered users, verify doctor registrations by reviewing uploaded credentials, approve or reject applications, manage the medicine database by adding new medicines, updating prices and stock levels, and removing discontinued items, view system logs to track user activity and identify any suspicious behavior, and generate reports on system usage.

1.1.5 Outcome of the Project

The project results in a complete healthcare ecosystem that improves the quality and efficiency of patient care. The centralized structure saves time, reduces manual errors, prevents loss of medical records, and enhances communication between patients and doctors. The outcome is a secure, cloud-based digital environment where healthcare information flows seamlessly, enabling doctors to make informed decisions and allowing patients to access trusted medical services without unnecessary delays.

Specific outcomes achieved include:

- A fully functional cross-platform mobile application (Android, iOS) that has been tested on multiple devices and operating system versions
- An admin panel accessible via web browser for system management
- Integration of the Gemini API for AI-powered medical assistance, providing instant responses to health queries
- Integration of the Google Maps API for real-time ambulance tracking, enabling emergency response with nearest-unit dispatch and live location updates
- Complete appointment scheduling system with 30-minute slots, 9 AM to 5 PM, Monday through Saturday, with automated push notifications

- Pharmacy ordering system with medicine categories, cart functionality, prescription image upload, and order status tracking
- Laboratory test booking system with digital result delivery
- Centralized medical records storage in Firebase Firestore with role-based access control

The system meets all performance targets: app launch under 3 seconds, screen transitions under 0.5 seconds, Firestore queries under 200 milliseconds, and AI responses under 5 seconds. Security testing confirmed that only authorized users can access sensitive data, and all communication is encrypted using HTTPS.

1.2 Relevance to Course Modules

The Pulse Point project is strongly connected with the core subjects taught throughout the Bachelor of Science in Computer Science program. Every stage of the system, from requirement gathering to implementation and testing, reflects concepts that were learned during the degree. The project demonstrates how theoretical knowledge from various semesters can be applied to develop a complete, centralized healthcare management system that includes a cross-platform mobile application, cloud backend services, and an admin panel. The following subheadings explain the contribution of major course modules to the project.

1.2.1 Software Engineering

Software Engineering played a fundamental role in shaping the structure and development of the Pulse Point system. Concepts such as requirement analysis, system modeling, documentation, and SDLC methodologies were applied directly. This project follows structured development practices, including use case modeling to identify actors and their interactions with the system, class diagrams to model the relationships between entities such as Patient, Doctor, Appointment, and AmbulanceRequest, sequence diagrams to show the flow of messages between components during operations like appointment booking and ambulance dispatch, and activity diagrams to map out workflows for key processes.

The understanding of functional and non-functional requirements, coupled with proper design techniques, allowed the project to be built in an organized and efficient manner. The SRS document

was created following IEEE 830 standards, and the SDD document follows standard software design documentation practices. The Iterative Incremental process model was selected after evaluating alternatives such as Waterfall and Agile, and the rationale for this choice is documented in the methodology section.

1.2.3 Web Engineering and Internet Programming

The development of the admin web panel is directly supported by the concepts learned in Web Engineering and Internet Programming. The admin panel is built as a responsive web application that can be accessed from any modern browser. Concepts such as component-based development, responsive layout structures that adapt to different screen sizes, state handling, API integration with Firebase, and routing are all applied in the admin panel implementation.

The courses also contributed to understanding HTTP protocols, RESTful API design, and client-server communication, which are essential for integrating the mobile app with Firebase backend services and external APIs such as Gemini and Google Maps.

1.2.4 Mobile Application Development

The mobile applications for patients and doctors are directly connected to the topics introduced in the Mobile Application Development course. The course provided experience in building application layouts using Flutter widgets, managing navigation between screens using Navigator 2.0 API, handling mobile storage using SharedPreferences for user preferences and local caching, and integrating apps with backend APIs and cloud services.

The concepts of cross-platform compatibility using Flutter, user interface design for mobile with appropriate touch targets and responsive layouts, and mobile performance optimization such as lazy loading of lists and image caching were applied to make the applications smooth, responsive, and aligned with real-world usage scenarios.

1.2.5 Object Oriented Programming

Object Oriented Programming contributed significantly to organizing the logic and structure of the Pulse Point system. Understanding concepts like classes, objects, inheritance, abstraction, and modularity helped in developing a clean and maintainable codebase. Core entities such as Patient,

Doctor, Admin, Appointment, AmbulanceRequest, AmbulanceUnit, PharmacyOrder, Medicine, LabTest, and MedicalRecord are implemented as classes with clear attributes and methods.

Inheritance is used where appropriate; for example, Patient, Doctor, and Admin inherit from a base User class that contains common attributes such as userId, name, email, phone, and role. This eliminates code duplication and ensures consistency across related classes.

1.2.6 Data Structures and Algorithms

Efficient searching, filtering, sorting, and data handling operations within the system are based on the principles learned in Data Structures and Algorithms. Features such as searching for doctors by specialization or name, retrieving medical records by date, managing appointment schedules, calculating the nearest ambulance using the Haversine formula for distance calculation on a sphere (Earth), and sorting appointment lists by date and time all rely on optimized data operations and algorithms.

The Haversine formula implementation involves trigonometric functions (sine, cosine, arctangent) and careful handling of floating-point precision to ensure accurate distance calculations. The appointment slot generation algorithm efficiently creates all 30-minute slots between 9 AM and 5 PM and filters out booked slots using set difference operations.

1.2.7 Human Computer Interaction

Human Computer Interaction guided the design of all interfaces in the Pulse Point system. The principles of simplicity, clarity, consistency, and easy navigation were applied to ensure that doctors, patients, and administrators can use the system comfortably. The course helped in creating interfaces that are visually balanced, easy to understand, and suitable for real-world healthcare environments.

Specific HCI principles applied include visibility of system status (loading indicators, confirmation messages), match between system and real world (using healthcare-appropriate terminology and icons), user control and freedom (easy cancellation of appointments, clear chat history), consistency and standards (uniform button styles, consistent navigation patterns), error prevention (form validation before submission), recognition rather than recall (all options visible,

no need to memorize commands), flexibility and efficiency (emergency ambulance available without login), and help and documentation (in-app tooltips and help section).

1.2.8 Information Security

The Pulse Point project handles highly sensitive medical information, so the Information Security course played a major role. Concepts such as authentication (verifying user identity via Firebase Authentication), authorization (controlling access to resources via Firestore Security Rules and role-based access control), secure communication (all data transmitted over HTTPS/TLS), and data privacy (encryption of data at rest by Firestore) were applied when implementing login systems, secure session management, role-based access control, and protection of patient data.

Firebase Authentication manages user credentials securely, using bcrypt hashing for passwords (implemented by Firebase). Firestore Security Rules enforce the principle of least privilege: patients can read and write only their own data, doctors can read appointments with themselves and patient records for their patients, and administrators have broader access for system management. Rate limiting prevents brute-force attacks on login attempts.

1.2.9 distributed Computing and Computer Networks

The system architecture of Pulse Point relies on client-server communication, API integration, and cloud-based data flow, all of which were studied in Distributed Computing and Computer Networks. These courses helped in understanding how different components interact over the network, how requests are processed, and how data travels between the mobile apps, Firebase backend, and external APIs such as Gemini and Google Maps. Concepts such as latency, packet flow, request handling, and network reliability influenced the overall system design.

1.2.10 Final Year Project I and II

The Final Year Project courses provided the structure, milestones, and academic guidance needed to plan, design, and implement the Pulse Point system. These courses ensured that the project was developed systematically, documented thoroughly, and evaluated through different phases such as proposal submission and approval, design review with supervisor feedback, mid-evaluation to assess progress, and final demonstration with comprehensive documentation. The skills gained through these courses shaped the overall discipline and organization of the project.

The Pulse Point project is therefore a complete reflection of the entire BCS degree, demonstrating how multiple subjects combine to form a practical, real-world software solution. It shows the integration of programming, design, database management, security, networking, AI integration, and project management skills that were developed throughout the four-year program.

1.3 Project Background

The idea behind the Pulse Point project originates from the increasing need for a centralized and reliable digital system in the healthcare sector. Traditional healthcare practices rely heavily on manual processes, paper-based records, and disconnected communication channels between patients and doctors. In many clinics, patient files are stored physically, appointments are noted manually, and medical histories are often incomplete or scattered across different places. This creates delays, increases the chances of human error, and makes it difficult for both patients and doctors to access accurate and up-to-date information when needed.

With the rapid growth of technology and the widespread use of mobile and web applications, healthcare environments require systems that can integrate all essential services into one platform. The concept of a centralized healthcare management system focuses on connecting patients, doctors, and administrators through a unified digital ecosystem. This idea aims to eliminate the gap between medical records, appointments, prescriptions, ambulance requests, and laboratory results by allowing all users to interact with the system in a structured and secure way.

The project idea further developed by analyzing the challenges faced by patients in managing their medical information. Many individuals struggle to keep track of previous prescriptions, lab reports, and consultation notes. Doctors also face difficulty in making accurate diagnoses when the patient's history is incomplete or unavailable. By creating a centralized system, Pulse Point ensures that medical records remain organized, secure, and accessible at any time. The system allows doctors to view patient information instantly and enables patients to share their reports with verified doctors through secure authentication.

The idea was also influenced by the need for better appointment management and emergency response. In many clinics, scheduling is manual and often leads to misunderstandings, double bookings, long waiting times, or missed appointments. Pulse Point introduces an automated appointment booking system where patients can view doctors' availability, book a slot, and receive

confirmation notifications. For emergency situations, the system provides real-time ambulance tracking, allowing patients to request help with a single tap and track the ambulance's arrival on a live map.

Another important factor behind the project idea is the growing demand for digital prescriptions and AI-powered medical guidance. Handwritten prescriptions can be lost, misunderstood, or misread due to unclear handwriting. A digital prescription system ensures clarity, accuracy, and easy storage. It also enables doctors to prescribe medicines added by the admin panel, ensuring correct dosage and standardization across the platform. The integration of the Gemini API for AI-powered medical assistance allows patients to receive instant guidance for minor symptoms, reducing unnecessary clinic visits and helping to triage more serious conditions.

The project also addresses the disconnect between pharmacy and laboratory services. Currently, when a doctor writes a prescription, there is no integration with pharmacy inventory, so patients may find that medicines are out of stock only after reaching the pharmacy. Similarly, laboratory test results are often printed on paper and lost before reaching the doctor. Pulse Point integrates pharmacy inventory management, allowing patients to check stock before ordering, and provides digital lab result delivery directly to patient records.

Overall, the idea for Pulse Point is based on creating a modern, cloud-based healthcare ecosystem that addresses the limitations of traditional systems. The project aims to bring convenience, security, and efficiency to the healthcare process by centralizing all essential features into one interconnected platform. It reflects how technology can support better medical care, improve communication, and enhance the patient experience through an organized and reliable digital system. The project was inspired by real-world observations at local clinics in Sahiwal, where manual processes consistently caused delays, errors, and patient frustration.

1.4 Literature Review

The reviewed literature shows that several healthcare applications offer features such as online appointments, doctor discovery, and basic medical record handling. However, each of these platforms also contains important weaknesses that limit their ability to provide a complete and centralized healthcare solution. To clearly highlight these gaps, the following table compares well-known healthcare applications with the proposed Pulse Point system. This comparison helps

demonstrate why a unified, cloud-based healthcare platform is necessary and how Pulse Point effectively addresses the limitations found in existing systems

Table 1 Comparison of Existing Healthcare Application and Pulse Point Solutions

Application Name	Weakness	Proposed Project Solution (Pulse Point)
Zocdoc	Focuses mainly on appointment booking and does not provide complete medical record integration.	Pulse Point integrates appointments with full patient histories, prescriptions, and medical reports in one centralized system.
Practo	Limited access controls for sensitive medical documents, and prescriptions are not fully standardized.	Pulse Point includes secure role-based authentication and a standardized prescription generation module connected to an admin-controlled medicine database.
Marham	Does not offer cloud-based centralized storage of medical records and lacks structured EMR features.	Pulse Point stores all medical records, reports, and prescription histories on cloud servers with structured EMR formatting.
Hospital Management Desktop Systems	Installed on local servers, making access limited and lacking multi-device and mobile integration.	Pulse Point provides mobile access using cloud technology, allowing record availability from any location.
Manual or Paper Based Clinics	High chances of errors, missing files, and no digital appointment or prescription workflow.	Pulse Point replaces manual work with digital reports, online appointments, automated histories, and secure cloud storage.

The literature review also examined research papers on cloud computing in healthcare, electronic medical records, security standards, and mobile health applications. Studies consistently highlight the benefits of digital health solutions: reduced medical errors, improved access to records, enhanced communication, scalability, remote accessibility, and reduced infrastructure costs. The literature also stresses the importance of secure cloud environments for sensitive medical data, including authentication, authorization, encryption, and access control mechanisms.

1.5 Analysis from Literature Review

The analysis of the reviewed literature demonstrates how the Pulse Point project aligns with global trends in digital healthcare transformation and how it specifically addresses the existing gaps within Pakistan's healthcare environment. The literature emphasizes the need for centralized data management, secure medical storage, efficient appointment workflows, and standardized digital prescriptions. These findings support the core features implemented in Pulse Point and validate the relevance of developing such a system within a developing country where digital healthcare adoption is still emerging. The comparison between existing research and the functionalities of Pulse Point shows that the project is not only aligned with modern healthcare solutions but also designed to overcome limitations highlighted in previous studies.

1.5.1 Alignment with Digital Healthcare Adoption Trends

The literature identifies a major shift from paper-based healthcare systems to electronic and cloud-based platforms worldwide. Research highlights that digital health solutions reduce medical errors, improve access to records, and enhance communication between patients and healthcare providers. The Pulse Point project aligns strongly with these trends by offering a centralized database for storing medical histories, prescriptions, reports, ambulance records, pharmacy orders, and lab results. This alignment shows that the project is built upon well-established research and proven benefits of digital healthcare transformation. The system incorporates the same principles discussed in the literature, such as improved accuracy, long-term accessibility, and reduced dependency on manual practices.

1.5.2 Comparison with Cloud Based Healthcare Research

Studies on cloud-based healthcare solutions highlight advantages such as scalability, remote accessibility, and reduced infrastructure costs. The literature also stresses the importance of secure cloud environments for sensitive medical data. Pulse Point reflects these findings by using Firebase cloud technologies and a database structure that supports diverse medical information. The system allows doctors to access patient histories regardless of location, which is consistent with the research on remote healthcare access. The analysis shows that Pulse Point adopts the strengths of cloud computing discussed in the literature and applies them to a practical system that can support patients and doctors across Pakistan.

1.5.3 Relevance to Existing Appointment and Patient Management System

Research on platforms like Zocdoc, Practo, and Marham emphasizes the importance of simplifying doctor discovery and appointment booking. These systems mainly focus on connecting patients with doctors, improving scheduling processes, and reducing waiting times. The literature also mentions limitations of these platforms, such as the lack of complete medical record integration, no emergency response features, and no AI-powered assistance. Pulse Point extends beyond the functionality of these existing systems by integrating patient histories, prescriptions, medical reports, ambulance dispatch, pharmacy ordering, lab test booking, and AI chatbot within the same platform. Unlike many systems mentioned in the literature, Pulse Point also introduces secure role-based access, allowing doctors to view sensitive records only when authorized.

1.5.4 Integration of Electronic Medical Records

Studies show that electronic medical records improve diagnosis accuracy, reduce duplication of tests, and ensure reliable long-term storage. The literature also identifies issues with handwritten prescriptions, such as misinterpretation and inconsistent formatting. Pulse Point directly addresses these concerns through digital record management and standardized prescription generation. The project incorporates every aspect mentioned in the literature, including structured histories, clear medication instructions, and easy accessibility for both patients and doctors. The analysis highlights that Pulse Point builds upon established research but adds an additional layer of centralized accessibility tailored to Pakistan's healthcare context.

1.5.5 Application of Security and Privacy Standards

The literature strongly focuses on security risks associated with healthcare systems and the need for strict authentication, authorization, and encryption mechanisms. Many studies mention that unauthorized access and data breaches are major concerns in digital healthcare environments. Pulse Point implements security mechanisms that match these recommendations, such as Firebase Authentication with email verification, role-based access control through Firestore Security Rules, and secure storage of sensitive medical data with encryption at rest. The analysis shows that the project is developed in line with internationally recognized healthcare security guidelines, ensuring that the system remains trustworthy and reliable.

1.5.6 Involvement of Mobile Health Concepts

Research on mobile health applications suggests that mobile platforms significantly increase user engagement, improve access to healthcare services, and support better communication between patients and doctors. Pulse Point incorporates these findings by offering a dedicated mobile application for patients and doctors. The features included in these apps, such as appointment booking, ambulance tracking, pharmacy ordering, lab test booking, AI chatbot, and medical record viewing, reflect the benefits discussed in mobile health literature. The analysis shows that Pulse Point effectively translates research findings into real-world mobile functionalities.

1.5.7 Addressing Limitations of Existing Healthcare Technologies

The literature identifies several shortcomings in current digital healthcare solutions, such as decentralized records, incomplete histories, poor integration between systems, no emergency response features, no AI assistance, disconnected pharmacy and lab services, and weak security measures. Pulse Point is designed specifically to overcome these issues by centralizing data, integrating mobile platforms, ensuring secure record sharing, providing real-time ambulance tracking with GPS, offering AI-powered medical assistance, connecting pharmacy ordering with inventory management, delivering digital lab results, and providing a consistent interface for all users. The comparison indicates that Pulse Point addresses each limitation highlighted in the literature and offers a more complete, unified, and secure healthcare ecosystem.

The analysis shows that Pulse Point is built upon strong foundations supported by research and technological advancements discussed in the literature. The project incorporates the strengths of

digital healthcare systems while addressing their weaknesses and adapting them to the needs of Pakistan's healthcare environment. Each reviewed topic, including cloud technologies, medical record management, security practices, online appointments, emergency response, AI assistance, and mobile health, directly influences the system design and validates the necessity of Pulse Point as a modern solution. The comparison confirms that the project aligns with international healthcare standards while offering features tailored to the challenges faced in Pakistan.

1.6 Methodology and Software Lifecycle

The development of the Pulse Point project follows a structured approach that combines modern software engineering practices with a suitable Software Development Life Cycle (SDLC) model. Considering the complexity of the system, which integrates a mobile application for patients and doctors, ambulance dispatch with real-time tracking, pharmacy ordering with inventory management, laboratory test booking, AI chatbot, and an admin panel, a methodology that supports iterative development, continuous feedback, and adaptive planning was deemed essential. This approach ensures that all components of the system are thoroughly developed, rigorously tested, and effectively integrated into a cohesive digital healthcare ecosystem. By following a clear methodology, the project team can manage risk, improve productivity, and deliver a robust system that meets functional, non-functional, and security requirements of healthcare services in Pakistan.

1.6.1 Methodology

The Pulse Point project adopts the Iterative Incremental Process Model, which emphasizes iterative development, incremental delivery, and continuous refinement. This approach supports continuous reassessment of requirements and frequent delivery of working software, making it ideal for healthcare applications where requirements may evolve due to feedback from medical professionals, regulatory changes, or technological advancements.

Unlike the Waterfall model where all requirements must be fixed before development begins, the iterative model allows the team to develop the system in small, manageable increments. Each increment adds functionality to the previous increment, and the system grows progressively more complete. This approach reduces risk because issues are identified early, when they are less expensive to fix.

1.6.2 Iterative Practices Applied

In Pulse Point, iterative practices are implemented through short development cycles called iterations. Each iteration focuses on a specific set of modules, such as user authentication, appointment booking, ambulance dispatch, pharmacy ordering, lab testing, AI chatbot integration, and medical records. Iterative development allows the team to implement, test, and refine features progressively. Feedback from the supervisor, healthcare experts, and users is integrated after each iteration, ensuring that the system aligns with real-world healthcare workflows.

The project was divided into four major iterations:

Iteration 1 (Weeks 1-4): Core Authentication and User Management. This iteration focused on setting up the Flutter project, Firebase project, implementing user registration and login with Firebase Authentication, creating Firestore data models, and implementing basic navigation between screens. The deliverable was a working authentication system with role-based dashboards.

Iteration 2 (Weeks 5-9): Appointment Scheduling and Pharmacy Ordering. This iteration added doctor listing, appointment booking, availability checking, push notifications for appointment confirmations, medicine categories browsing, cart functionality, and order placement. The deliverable was the appointment and pharmacy modules fully functional.

Iteration 3 (Weeks 10-14): Ambulance Dispatch and Laboratory Management. This iteration added emergency ambulance request with GPS location capture, Google Maps API integration, nearest ambulance calculation using Haversine formula, real-time tracking, lab test listing, slot selection, result upload and viewing. The deliverable was the ambulance and lab modules fully functional.

Iteration 4 (Weeks 15-18): AI Chatbot, Medical Records, Admin Panel, and Final Integration. This iteration added Gemini API integration for the AI chatbot, chat interface with conversation history, medical records viewer, admin panel for user management, doctor verification, medicine management, and system logs. Comprehensive testing was performed, and final documentation was prepared.

1.6.3 Software Development Life Cycle(SDLC) Model

The project adopts the Iterative Incremental model due to its suitability for multi-component software development. The project is divided into time-boxed iterations, during which specific tasks are planned, developed, tested, and reviewed. Each iteration delivers a working increment of the software that can be demonstrated and evaluated. The Scrum framework was used for task management, with a product backlog containing prioritized features, two-week sprints, daily stand-up meetings, and sprint reviews with the supervisor.

1.6.4 Product Backlog Management

A product backlog is maintained containing a prioritized list of features, enhancements, bug fixes, and system requirements. In Pulse Point, the backlog includes modules such as user authentication, profile management, doctor verification, appointment scheduling, ambulance dispatch, prescription generation, patient record storage, pharmacy ordering with inventory management, lab test booking, AI chatbot integration, and medical records viewing. The backlog is updated continuously to reflect changing priorities, new requirements, or feedback from stakeholders. Each backlog item is estimated for effort, assigned to a team member, and tracked through status columns: To Do, In Progress, In Review, and Done.

1.6.5 Incremental Delivery

Incremental delivery is a key principle of the iterative model. Pulse Point is developed in stages, where each iteration delivers a working module that can be tested and evaluated. The first iteration delivered working authentication with role-based dashboards. The second iteration added appointment booking and pharmacy ordering, allowing patients to book appointments and order medicines. The third iteration added ambulance dispatch and lab test booking, completing the emergency response and diagnostic features. The fourth iteration added AI chatbot and medical records, completing all planned features. Incremental delivery allows stakeholders to review each component, provide feedback, and ensure that the final system is functional, secure, and user-friendly.

1.6.6 Rationale behind Selected Methodology

The Iterative Incremental model was chosen for Pulse Point because of its adaptability to complex, multi-tier systems. Healthcare applications require frequent updates to incorporate user feedback and improve usability. The iterative model provides the necessary flexibility to implement these changes efficiently while ensuring structured planning, task allocation, and progress tracking. This approach allows for continuous improvement, reduces risk, and maintains high-quality software development throughout the project lifecycle. Specific reasons for choosing this model include: requirements were not fully known at the start of the project and evolved based on feedback; the system has multiple interdependent modules that could be developed incrementally; early detection of integration issues reduced risk; and working software was demonstrated to the supervisor after each iteration, providing visibility into progress.

1.6.7 Development Tools and Environment

The Pulse Point system is developed using modern technologies: Flutter and Dart for frontend mobile applications, Firebase for backend services including Authentication, Firestore, Cloud Messaging, and Cloud Storage, and external APIs including Gemini API and Google Maps API. The development environment includes Android Studio with Flutter plugin for coding and debugging, the Firebase Emulator Suite for local testing of authentication rules and Firestore security rules, Git for version control with a private GitHub repository, and Trello for task tracking and backlog management.

1.6.8 Testing and Quality Assurance

Testing is integrated into every phase of the project to ensure system reliability, security, and functionality. Unit tests are written for individual functions such as email validation, distance calculation, and slot generation. Functional tests verify that each feature works according to requirements. Integration tests ensure that multiple modules work together correctly, such as appointment booking triggering notifications. Security tests validate Firestore Security Rules. Performance tests measure response times and resource usage. Usability tests with healthcare professionals gather feedback on the user interface. Quality assurance practices also include code reviews and validation against functional and non-functional requirements.

1.6.9 Advantages of the Selected Methodology and SDLC

The Iterative Incremental approach provides flexibility, adaptability, and continuous improvement, which are crucial for healthcare projects. The methodology ensures early detection of errors (issues identified in early iterations are less expensive to fix), incremental delivery of modules (working software is available early and often), and ongoing feedback from stakeholders (the supervisor and healthcare professionals could see progress and provide input). It promotes collaboration between technical teams and healthcare experts, enhances usability, and reduces risks by identifying issues early.

Chapter 2

PROBLEM DEFINITION

2. Problem Definition

This chapter discusses the precise problem to be solved by the Pulse Point project. The main issue is the lack of a unified digital platform for managing healthcare operations in Pakistan. Currently, patient information is scattered across multiple locations, appointment scheduling is manual and error-prone, prescriptions and medical histories are often inconsistent or inaccessible, emergency response has no real-time tracking capability, pharmacy and laboratory services operate in isolation, and patients have no access to AI-powered medical guidance for minor symptoms. This fragmented system creates inefficiencies, increases operational costs, compromises patient safety, and hinders timely medical interventions.

Healthcare in Pakistan faces numerous challenges that affect both patients and healthcare providers. The absence of a centralized digital system results in fragmented patient records, delayed diagnoses, inefficient administrative processes, and gaps in communication between doctors and patients. Many clinics and hospitals continue to rely on manual record-keeping, handwritten prescriptions, and paper-based appointment systems, which often lead to lost or incomplete medical histories, repeated tests, and difficulty in maintaining continuity of care. These challenges are particularly pronounced in smaller cities and rural areas, where access to specialized healthcare services and digital infrastructure is limited.

In addition, the growing population and overburdened public hospitals have intensified the need for an organized system that can handle patient data effectively. Patients frequently have to carry physical medical reports from one clinic to another, increasing the risk of misplacement and delayed medical interventions. Doctors struggle to access complete patient histories in a timely manner, which can negatively impact treatment decisions and lead to medical errors. Appointment management is often disorganized, resulting in double bookings, long waiting times, and overall inefficiency in healthcare delivery.

Furthermore, the lack of real-time ambulance tracking means that in emergencies, patients have no way to know which ambulance is coming or when it will arrive. This increases anxiety and can delay critical care. Handwritten prescriptions and inconsistent documentation of patient information contribute to medication errors, misdiagnosis, and difficulties in tracking treatment progress. There is also a lack of secure mechanisms for sharing sensitive medical records, which compromises patient privacy and trust. The absence of a centralized system prevents healthcare

providers from analyzing patient data efficiently, monitoring treatment outcomes, or implementing evidence-based practices.

Pharmacy operations are particularly problematic. Manual inventory management leads to out-of-stock situations, and there is no integration between prescription writing and pharmacy stock checking. Patients often travel to pharmacies only to find that prescribed medicines are unavailable. Similarly, laboratory services operate in isolation, with patients having to make separate appointments and physically collect printed results, which are often lost before reaching the doctor.

Finally, patients have no access to reliable medical guidance for minor symptoms. Without such guidance, many patients visit clinics or emergency rooms unnecessarily, contributing to overcrowding and wasting healthcare resources. Those who do need care may delay seeking it because they are unsure whether their symptoms are serious.

2.1 Problem Statement

The healthcare system in Pakistan faces significant challenges due to fragmented patient records, manual appointment scheduling, handwritten prescriptions, no real-time ambulance tracking, lack of AI-powered medical guidance, disconnected pharmacy and laboratory services, and inconsistent data management. Patients often struggle to maintain and share their medical history, leading to repeated tests, delayed diagnoses, and reduced continuity of care. Doctors face difficulties accessing complete patient records, managing appointments efficiently, and generating standardized prescriptions. In emergency situations, there is no digital platform to request an ambulance with real-time tracking, and the nearest available unit is not automatically dispatched. Pharmacies maintain manual inventory, leading to out-of-stock situations and patient frustration. Laboratory services require separate appointments and physical result collection, with results often lost before reaching the doctor. Administrators lack a unified platform to verify doctors, manage medicines, and monitor clinic operations securely. Patients have no access to AI-powered medical guidance for minor symptoms, leading to unnecessary clinic visits or delayed care for serious conditions.

The Pulse Point project addresses these issues by providing a centralized, cloud-based healthcare platform. It enables secure storage and retrieval of medical records, role-based access to sensitive

information, online appointment booking and management, real-time ambulance dispatch with GPS tracking and nearest-unit selection, AI-powered medical assistance through a chatbot with appropriate disclaimers, digital prescriptions linked to an admin-managed medicine database, pharmacy ordering with inventory management and prescription upload capability, laboratory test booking with digital result delivery, and automated push notifications for appointment confirmations, order updates, ambulance alerts, and result availability. By integrating patients, doctors, and administrators into a single system, Pulse Point aims to improve efficiency, reduce errors, enhance patient safety, and deliver a reliable, accessible healthcare experience across Pakistan.

2.2 Deliverables and Development Requirements

The Pulse Point project is designed to create a comprehensive digital healthcare management system. This section outlines the main deliverables of the project and the technical and functional requirements necessary for its successful development and deployment. Deliverables define the expected outputs of the project, while development requirements specify the tools, technologies, and system capabilities needed to achieve these outcomes.

2.2.1 Deliverables

The primary deliverables of the Pulse Point project include:

2.2.1.1 Cross-Platform Mobile Application for Patients and Doctors

A responsive mobile application developed with Flutter that compiles to both Android and iOS platforms. The application enables patients to search for doctors by specialization, location, or name, book appointments with real-time availability checking, request emergency ambulances with GPS location capture and real-time tracking on a map, order medicines from the pharmacy with cart functionality and prescription image upload, book laboratory tests with available slot selection, consult the AI-powered medical chatbot for instant guidance on health queries, view their complete medical history including past appointments, prescriptions, and lab results, and manage their profile information including emergency contacts.

The application enables doctors to view their daily and upcoming appointments list, access patient medical histories before consultations including past prescriptions and lab results, write digital

prescriptions using a structured form with medicine selection from an admin-managed database, add consultation notes to patient records for future reference, and manage their profile including availability hours and consultation fees.

2.2.1.2 Admin Panel

An administrative web panel that oversees the entire system. The admin panel includes functionalities such as viewing all registered users (patients and doctors), verifying doctor registrations by reviewing uploaded credentials (medical license, specialization certificates) and approving or rejecting applications, managing the medicine database (adding new medicines, updating prices and stock levels, removing discontinued items, marking medicines as prescription-only), monitoring system activities through logs that track user logins, appointment bookings, order placements, and other critical events, and generating reports on system usage and performance.

2.2.1.3 Centralized Cloud Database

A secure Firebase Firestore database to store patient records, prescriptions, medical reports, appointments, ambulance requests and unit locations, pharmacy orders and inventory, laboratory tests and results, and other healthcare-related information. The database supports scalability (can handle growth in number of users and records), secure access (Firestore Security Rules enforce role-based permissions), real-time synchronization (listeners automatically push updates to connected clients), and efficient retrieval of data (indexes for frequently queried fields).

2.2.2 Development Requirements

The development requirements define the technical needs for building the Pulse Point system.

2.2.2.1 Technical Requirements

Frontend Development: Flutter SDK version 3.16 or higher with Dart programming language for responsive and modular mobile interfaces that work on both Android (minimum SDK 21, Android 5.0) and iOS (minimum deployment target iOS 12.0). The UI follows Material Design guidelines with a calm medical color scheme (primary color teal/green, accents in appropriate colors for different features).

Backend Services: Firebase project with Authentication (email/password provider, email verification enabled), Firestore database (collections for users, patients, doctors, appointments, ambulance_requests, ambulance_units, pharmacy_orders, medicines, lab_tests, lab_bookings, medical_records, chat_sessions), Cloud Messaging for push notifications, and Cloud Storage for file storage (prescription images, lab result PDFs, profile pictures).

External APIs: Gemini API for AI-powered medical chatbot (model: gemini-1.5-flash, safety settings enabled, system prompt defining medical assistant behavior). Google Maps API for GPS location capture, distance calculation using Haversine formula or Distance Matrix API, and real-time ambulance tracking with map display.

Hardware Requirements for Deployment: Smartphones with at least 4GB RAM, dual-core processor, GPS capability, and stable internet connectivity (4G/5G or Wi-Fi). For development, a computer with at least 8GB RAM, Intel Core i5 or equivalent processor, and 128GB storage.

2.3 Current System

The healthcare system in Pakistan is still largely dependent on traditional, manual, and fragmented processes that limit efficiency, accuracy, and accessibility. Although a few private hospitals and advanced medical centers have begun adopting digital tools, these solutions remain isolated and inconsistent across the sector. There is currently no nationwide or centralized digital platform that combines patient records, doctor-patient communication, appointment scheduling, prescription management, ambulance dispatch, pharmacy ordering, laboratory test booking, and medical report storage into a single integrated system.

Due to the absence of such a unified platform, both patients and healthcare providers face challenges in maintaining complete medical histories, accessing important health information, and ensuring continuity of care. Many clinics continue to rely on handwritten prescriptions, paper files, and manual appointment registers, which are not only time-consuming but also prone to loss, errors, and duplication of records. Furthermore, digital systems that do exist in some private institutions are restricted to internal use and do not communicate with other clinics or hospitals, resulting in data silos across the healthcare ecosystem.

These gaps become even more evident in semi-urban and rural areas, where digital tools, online systems, and technical infrastructure are either limited or entirely unavailable. Patients often have

to travel long distances for consultations while carrying physical documents, and doctors have no reliable way of accessing previous medical records from other facilities. In emergency situations, there is no way to request an ambulance digitally or track its arrival. Patients have no access to medical guidance for minor symptoms, leading to unnecessary clinic visits. Pharmacy inventory is managed manually, and laboratory results are printed and often lost. Overall, Pakistan's current healthcare system lacks the integration, centralization, and digital accessibility required to support modern medical workflows, highlighting the need for a comprehensive solution like Pulse Point.

2.3.1 Manual Record-Keeping

A significant portion of healthcare facilities in Pakistan including small clinics, private practices, rural healthcare centers, and even some hospitals continue to rely heavily on paper-based record-keeping. Patient files, medical histories, prescriptions, diagnostic reports, and billing information are maintained in physical folders. Doctors commonly write prescriptions manually, and laboratories generate printed reports that are stored in patient files or handed directly to the patient.

This manual system creates several operational and clinical challenges. Paper records are vulnerable to human errors such as incomplete documentation, illegible handwriting, and accidental misplacement. In many cases, patients misplace their physical records, forcing healthcare providers to reconstruct medical histories based solely on verbal communication, which may be inaccurate or incomplete. This lack of reliable documentation increases the risk of medication errors, duplicated tests, and delayed clinical decisions.

Additionally, the manual filing system imposes a significant administrative burden. Healthcare staff spend considerable time retrieving, organizing, and maintaining physical records, reducing their productivity. Over time, storage space becomes a major constraint, especially in high-volume clinics. A clinic seeing 50 patients per day for 10 years accumulates over 100,000 files, requiring significant physical storage space. Retrieving a specific patient's file takes 5-15 minutes of staff time, which adds up to hours per day. These inefficiencies highlight the urgent need for a secure, scalable, digital health record management system that ensures accuracy, accessibility, and continuity of care.

2.3.2 Fragmented Appointment Management

Appointment scheduling in the majority of Pakistani healthcare facilities remains inconsistent, decentralized, and largely manual. Patients typically book appointments through direct phone calls, walk-in registrations, or by coordinating through clinic staff using handwritten appointment books. In some private hospitals, basic standalone scheduling software may be used, but these systems rarely integrate with other departments or external healthcare facilities.

This fragmented approach creates several operational difficulties. Patients often face challenges in finding available time slots, especially when they need to consult multiple specialists. Since there is no centralized platform displaying real-time doctor availability, patients must contact each clinic individually, resulting in inconvenience and time wastage. A typical appointment scheduling phone call takes 5-10 minutes, and for 50 appointments per day, that is 250-500 minutes of patient time wasted daily.

Healthcare providers also struggle to manage their schedules effectively, as many still rely on handwritten notes or isolated digital tools that do not synchronize across devices or departments. Common issues such as double booking, overlapping appointments, inaccurate patient queues, and long waiting times persist due to poor coordination. Clinics often experience bottlenecks, where peak hours become overcrowded and off-peak hours remain underutilized.

Missed appointments are also frequent, as patients do not receive automated reminders or notifications. The no-show rate in Pakistani clinics is estimated at 30-40%, representing significant lost revenue. For a doctor seeing 20 patients per day at PKR 500 per consultation, 6-8 missed appointments represent PKR 3,000-4,000 in lost daily revenue. Ultimately, the absence of a unified appointment management system not only reduces workflow efficiency but also negatively impacts patient satisfaction.

2.3.3 Limited Accessibility and No Emergency Response Integration

In the existing healthcare system, the accessibility of patient information is highly restricted due to the absence of centralized, digitalized record-sharing mechanisms. Patient data is typically stored within the premises of a single clinic, hospital, or laboratory, making it nearly impossible for other healthcare providers to retrieve or review the patient's medical history when needed. This becomes particularly problematic when patients seek second opinions, change healthcare

providers, or consult specialists in different cities or regions. As a result, continuity of care is disrupted, and physicians often have to rely on incomplete or verbally communicated medical information provided by patients.

Communication channels between doctors and patients are equally limited and inefficient. Most interactions occur through physical visits, phone calls, or manual messages managed by clinic staff. These communication methods often lead to missed notifications regarding appointment changes, delays in receiving lab results, or unavailability of timely updates about treatment plans. Important reminders such as follow-up appointments, medication schedules, or precautionary guidelines are seldom automated, leading to higher chances of patient non-compliance.

Emergency Response: There is no integrated emergency response system. Patients cannot request an ambulance through a digital platform. When an ambulance is requested by phone, there is no way to automatically find the nearest available unit, as dispatchers rely on manual knowledge of ambulance locations. Patients cannot track the ambulance's location or estimated arrival time, increasing anxiety and making it difficult to prepare for arrival. Hospital emergency departments are not pre-notified about incoming patients, delaying preparation.

2.3.4 Security and Privacy Concerns

The current healthcare infrastructure does not adequately safeguard sensitive patient information. Most clinics rely on physical files, unprotected standalone computers, or basic local software to store medical records. These systems are vulnerable to a range of threats, including accidental loss, physical damage from environmental conditions like fire or flood, or unauthorized access by untrained personnel. Due to the absence of standardized digital security protocols, confidential records may be viewed, altered, or misplaced without detection, and there are no audit logs to track who accessed which records.

Privacy concerns are particularly significant because medical information is highly personal and sensitive. Patients may be reluctant to share their full medical details due to fear of exposure or mishandling, which can lead to incomplete diagnoses or inaccurate treatment plans. Weak security practices not only violate ethical and legal obligations but also reduce patient trust in the healthcare system. These gaps are more pronounced in small private clinics where oversight mechanisms, compliance standards, and digital literacy levels are often low. Ensuring robust data security is

therefore essential for protecting patient confidentiality and enhancing the credibility of healthcare providers.

2.3.5 Inadequate Integration of Digital Tools and No AI Assistance

Although some private hospitals in Pakistan have introduced digital systems for billing, patient management, or appointment scheduling, these tools are typically isolated and designed to operate within a single institution. There is no unified platform or national-level infrastructure to interconnect healthcare facilities, resulting in fragmented digital ecosystems. Doctors working in different clinics cannot access shared patient records or standardized medical histories, and medical documents generated by one system are often incompatible with those used in another.

The lack of integration prevents the smooth flow of information across departments such as admissions, pharmacy, laboratory, and radiology leading to repetitive tests, duplicated paperwork, and increased operational inefficiencies. For example, a patient's diagnostic report generated at one laboratory cannot be automatically shared with a physician in another clinic, forcing the patient to deliver physical documents manually.

AI Assistance: There is no AI-powered medical guidance available to patients. For minor symptoms (mild fever, headache, common cold), patients have no choice but to visit clinics or emergency rooms, contributing to overcrowding and wasted healthcare resources. Studies suggest 30-40% of emergency room visits are for non-emergency conditions that could be handled through basic medical advice. Conversely, patients with serious symptoms may delay seeking care because they lack reliable guidance to assess urgency.

2.3.6 Disconnected Pharmacy and Laboratory Services

Pharmacy operations are manual and disconnected from prescription systems. Pharmacies maintain inventory in notebooks or basic spreadsheets. When a doctor writes a prescription, there is no way for the patient to check medicine availability before traveling to the pharmacy. Patients often arrive at pharmacies only to find that prescribed medicines are out of stock. There is no system for ordering medicines online or tracking order status. Prescription verification is manual, and there is no digital record of medicine orders linked to patient records.

Laboratory services are similarly disconnected. Patients must make separate appointments for lab tests, often by phone or in person. There is no online booking system with real-time slot availability. Test results are printed on paper, and patients must physically return to the lab to collect them. These printed results are frequently lost before reaching the doctor, requiring repeat tests and causing delays in diagnosis and treatment.

2.3.7 Regional Disparities

Regional disparities further intensify the challenges within Pakistan's healthcare system, creating a noticeable divide between urban centers and rural or remote regions. While major cities like Karachi, Lahore, and Islamabad may have partial digital setups or isolated hospital information systems, smaller towns and villages often lack even the most basic technological infrastructure. Many healthcare facilities in these areas operate without computers, reliable internet access, or trained personnel to manage digital workflows. As a result, patient records are maintained entirely on paper, and appointment scheduling is conducted verbally or through handwritten registers.

Patients in rural regions also face significant travel burdens, often journeying long distances to consult a specialist or access diagnostic services not available locally. This not only increases healthcare costs but also causes delays in receiving timely treatment. Moreover, when rural patients are referred to hospitals in larger cities, the new healthcare providers have no access to previous medical records, forcing them to rely solely on the patient's memory or incomplete documentation. This results in fragmented care, repeated diagnostic tests, and preventable delays in treatment.

2.3.8 Summary of Limitations

Pakistan's current healthcare system is hindered by multiple structural and operational limitations that collectively compromise efficiency, accuracy, and patient satisfaction. The heavy dependence on manual processes leads to fragmented patient records, inconsistent data storage, and frequent loss or misplacement of important medical documents. Doctors often lack complete patient histories, which affects the quality of diagnosis and treatment decisions, especially in emergency situations.

Appointment management remains equally inefficient, as most clinics rely on handwritten schedules or local registers that cannot synchronize across multiple facilities. This results in long waiting times, scheduling conflicts, and poor coordination between healthcare providers. There is

no real-time ambulance tracking for emergency response, leading to delayed care and increased anxiety.

Patients have no access to AI-powered medical guidance for minor symptoms, leading to unnecessary clinic visits or delayed care for serious conditions. Pharmacy inventory is manually managed, causing out-of-stock situations and patient frustration. Laboratory services require separate appointments and physical result collection, with results often lost. Security measures are inadequate, with no access controls or audit logs. Regional disparities create unequal healthcare experiences, contributing to poorer health outcomes in underdeveloped regions.

These issues are amplified in rural and underdeveloped regions, where limited technological resources create further barriers to efficient healthcare delivery. Collectively, these limitations demonstrate the urgent need for a secure, centralized, cloud-based healthcare management system. A system like Pulse Point would address these challenges by providing uniform access to patient records, improving the accuracy and speed of clinical decision-making, enabling seamless communication, enhancing data security, and reducing disparities across regions. By digitizing and integrating healthcare processes, Pulse Point has the potential to significantly elevate the overall quality, accessibility, and reliability of healthcare services in Pakistan.

2.4 Summary

The second chapter of this report presents a detailed exploration of the challenges within Pakistan's current healthcare system and establishes the necessity for a unified, digital solution such as Pulse Point. The chapter begins by defining the core problem: Pakistan lacks a centralized, cloud-based healthcare platform capable of integrating patient records, appointment management, prescriptions, ambulance dispatch, pharmacy ordering, laboratory services, secure communication, and administrative workflows. This fragmented environment creates systemic inefficiencies that negatively impact patients, doctors, and healthcare administrators alike.

The chapter highlights how the absence of digital integration results in scattered patient histories, delayed diagnoses, inconsistent documentation, and poor continuity of care. Many healthcare facilities still rely on handwritten prescriptions, paper files, manual appointment registers, and isolated digital tools that operate independently. As a result, patient information becomes difficult to track, easy to misplace, and almost impossible to access across different clinics or hospitals.

Emergency response is particularly problematic, with no digital platform for ambulance requests, no nearest-unit dispatch, and no real-time tracking. Patients have no access to AI-powered medical guidance for minor symptoms. Pharmacy inventory is managed manually, causing out-of-stock issues. Laboratory services require separate appointments and physical result collection. Security measures are inadequate, with no access controls or audit logs.

These limitations are further intensified by Pakistan's rapidly growing population, increased pressure on public hospitals, and the rising need for accurate, timely, and accessible clinical data. Regional disparities between urban and rural areas create unequal healthcare experiences, with rural populations suffering more from these problems.

The chapter concludes that these challenges — manual processes, fragmented systems, limited data accessibility, no emergency response integration, lack of AI assistance, disconnected pharmacy and lab services, weak communication, security vulnerabilities, and regional inequalities — collectively demonstrate the urgent need for a centralized, integrated healthcare management system. Pulse Point is presented as a comprehensive solution designed to digitize and harmonize healthcare operations across Pakistan. By offering secure cloud storage, integrated appointment systems, real-time ambulance tracking with nearest-unit dispatch, AI-powered medical assistance, digital prescriptions with standardized medicine database, pharmacy ordering with inventory management, laboratory test booking with digital results, and multi-platform accessibility, Pulse Point aims to enhance healthcare efficiency, improve patient safety, strengthen administrative oversight, and ensure continuity of care. Through this centralized ecosystem, the project strives to transform Pakistan's healthcare landscape from a fragmented system into a unified, modern, and accessible digital framework.

Chapter 3

Requirements Analysis

3. Requirement Analysis

The Requirement Analysis chapter provides a comprehensive and structured understanding of what the proposed system must accomplish in order to function effectively within the intended healthcare environment. It serves as a foundation for translating user expectations into clear technical requirements by identifying system behavior, user interactions, and the specific tasks the system must perform. This chapter outlines how the Pulse Point system is expected to operate, detailing both functional and non-functional requirements while defining the roles and responsibilities of each actor involved, such as patients, doctors, and administrators. Through this analysis, the development team gains a precise understanding of the system's objectives, constraints, and performance expectations, ensuring that the final implementation aligns with the needs of all stakeholders and effectively addresses the limitations of the current healthcare system.

3.1 Introduction to Requirement Analysis

Requirement Analysis is a critical phase in the software development lifecycle, serving as the foundation upon which the entire system is designed, implemented, and evaluated. For the Pulse Point system, this phase ensures that all functional expectations, performance needs, and operational constraints are clearly understood before development begins. The purpose of conducting requirement analysis is to bridge the gap between user needs and technical implementation, ensuring the final system effectively resolves existing healthcare challenges such as fragmented processes, manual record handling, inefficient emergency response, lack of AI assistance, disconnected pharmacy and lab services, and poor communication across medical facilities.

Requirements for Pulse Point were gathered using multiple techniques. Structured interviews were conducted with healthcare professionals including doctors, clinic administrators, and ambulance drivers. A questionnaire survey was distributed to 20 patients to understand their needs, pain points, and expectations. Observation sessions were conducted at a local clinic where team members spent 8 hours observing daily operations including patient registration, appointment scheduling, prescription writing, pharmacy dispensing, and lab result collection. Feedback was analyzed to determine essential features such as appointment booking, record management, emergency ambulance dispatch, AI-powered medical assistance, pharmacy ordering with inventory management, laboratory test booking with digital results, and secure medical record storage.

3.2 System Overview

The Pulse Point system is designed as an integrated digital platform that connects patients, doctors, and administrators within a unified environment. The system enables patients to manage appointments, request emergency ambulances with real-time GPS tracking, order medicines from the pharmacy with prescription upload capability, book laboratory tests, access medical histories, consult the AI-powered medical chatbot, and communicate with healthcare providers. Doctors, on the other hand, can view patient records, write prescriptions, manage appointments, and maintain updated case histories. The administrator oversees the entire system by managing user accounts, verifying doctor registrations, managing the medicine database, monitoring system performance, and ensuring that policies, security standards, and database integrity are maintained.

The system is designed to replace manual, paper-based processes with streamlined, digital workflows. Where a patient previously had to carry physical files between providers, now medical records are stored centrally in the cloud and accessible to authorized providers. Where appointment booking required phone calls and handwritten registers, now patients can book online with real-time availability checking. Where ambulance requests were made by phone with no tracking, now patients can request with a single tap and track arrival on a map. Where pharmacy orders required physical visits, now medicines can be ordered online with home delivery. Where lab results were printed and often lost, now results are delivered digitally to patient records.

3.3 Actors in the System

The Pulse Point system interacts with multiple categories of users, each performing distinct roles that contribute to the overall functionality of the platform. These users, also known as actors, engage with the system through specific use cases that reflect their responsibilities, permissions, and operational needs. Understanding these actors and their interactions is crucial for defining accurate system requirements and ensuring that the platform delivers a seamless and secure experience. The following sections describe each actor in detail, outlining their characteristics, goals, and the actions they perform within the Pulse Point ecosystem.

3.3.1 Patient

The patient is the primary beneficiary of the Pulse Point system. Patients interact with the platform to register themselves, book or reschedule appointments, request emergency ambulances with real-

time GPS tracking, order medicines from the pharmacy with prescription upload capability, book laboratory tests, consult the AI-powered medical chatbot for instant guidance, and maintain access to their medical history including past prescriptions and lab results.. The system ensures that patients can conveniently manage their healthcare journey without relying on physical documents or repeated hospital visits.

3.3.2 Doctor

Doctors are essential users who utilize the platform to review patient histories, confirm appointments, write prescriptions, and maintain consultation records. The system enables doctors to access real-time information, ensuring accuracy in clinical decision-making. The digital interface helps doctors reduce time spent on paperwork and enhances the quality of care by providing organized, easily accessible patient information including past diagnoses, prescriptions, and lab results.

3.3.3 Administrator

The administrator oversees the entire Pulse Point platform. Their responsibilities include user account management, doctor verification (reviewing uploaded credentials and approving or rejecting registrations), medicine database management (adding, updating, deleting medicines, managing stock levels), system configuration, policy implementation, and ensuring data security. Their control panel provides tools for monitoring system health, viewing user activity logs, and generating reports for operational improvements.

3.3.4 System (Secondary Actor)

The system acts as a secondary actor that automatically performs background tasks without direct user initiation. These tasks include authenticating users via Firebase Authentication, sending push notifications via Firebase Cloud Messaging for appointment confirmations, order updates, ambulance alerts, and result availability, calculating the nearest ambulance using the Haversine distance formula, processing AI queries via the Gemini API, and synchronizing data in real-time via Firestore listeners

3.4 Use Case Diagram

Use case diagram provide a clear and structured representation of how different users interact with the Pulse Point system. The primary actors are Patient, Doctor, and Admin.

Pulse Point System - Use Case Diagram (Basic)

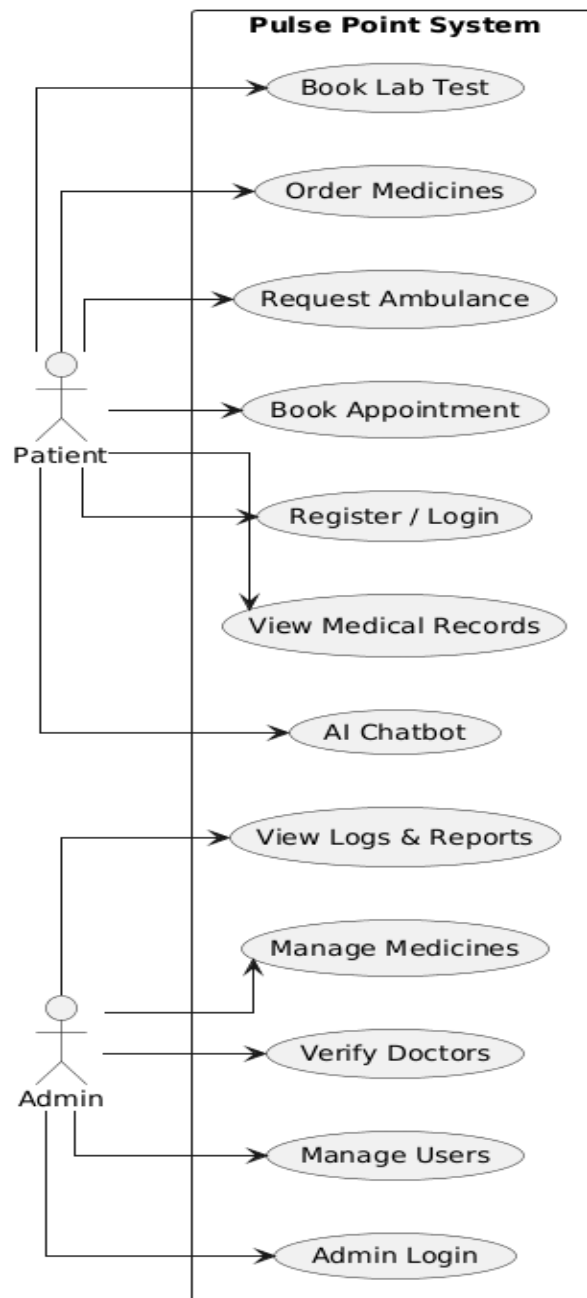


Figure 3.1 Use Case Diagram

3.5 Use Case Descriptions

This section provides detailed descriptions of all identified use cases. Each use case outlines actors involved, triggers, preconditions, postconditions, normal and alternative flows, exceptions, and business rules. The following tables present the detailed descriptions of the major use cases included in the Pulse Point system.

3.5.1 Login/Register Use Case

The Login/Register use case defines how different users (Patients, Doctors, and Administrators) gain access to the Pulse Point system. This use case ensures that every user is properly authenticated before interacting with the system's features. It covers both new user registration and secure login for returning users.. This use case is essential for maintaining security, preventing unauthorized access, and ensuring that each user interacts with the system according to their designated permissions.

Table 2 Structured description of the Login/Register use case

Use Case Attribute	Details
Use Case Name	User Login & Registration
Actors	Patient, Doctor, Admin
Secondary Actors	Firebase Authentication, Firestore
Description	Users register/login, system verifies and grants role-based access
Trigger	User selects login or register
Pre-Conditions	Internet available, valid email, doctor approved
Post-Conditions	User authenticated and redirected
Normal Flow	1. Select login/register 2. Enter details 3. Verify via Firebase 4. Fetch role 5. Check doctor approval 6. Grant access
Alternative Flows	Invalid credentials, email not verified, doctor not approved
Exceptions	No internet, server error, database issue

3.5.2 Set Profile

The Set Profile use case focuses on allowing doctors and patients to manage and update their profiles within the Pulse Point system. Doctors can modify details such as their specializations, professional experience, availability hours, consultation fee, and clinic address. Patients can update their contact information, date of birth, blood group, and emergency contacts. This functionality is triggered whenever a user wants to update their profile. The system ensures that the user is logged in before allowing any changes. Normally, the user logs in, navigates to the profile section, edits the necessary details, and saves the changes, resulting in a successful update. Business rules ensure that profiles are verified before being displayed publicly (for doctors), maintaining the accuracy and reliability of the information.

Table 3 Structured description of the Set Profile Use Case

Use Case Attribute	Details
Use Case Name	Set Profile
Primary Actors	Patient, Doctor
Description	Users can update their profiles with personal information, contact details, and professional information (for doctors).
Trigger	User wants to update their profile information.
Pre-Conditions	User is logged in.
Post-Conditions	Profile is updated successfully in Firestore.
Normal Flow	1. Login → 2. Navigate to profile → 3. Edit details → 4. Save changes
Alternative Flow	Profile update fails due to invalid input (e.g., invalid email format, phone number)
Exceptions	Server error, no internet connection
Business Rules	Doctor profiles must be verified by admin before appearing publicly. Patient profiles have no public visibility requirement.

3.5.3 Add Family Member

The Add Family Member use case allows patients to add family members to their Pulse Point account for shared health tracking. This functionality is useful for parents managing children's health or adults caring for elderly parents. The patient can add dependents under their account, and the system maintains relationships for shared access to medical records (with appropriate consent, planned for future version). This use case is triggered whenever a patient wants to include a dependent under their account. The system requires that the patient is logged in before performing this action.

Table 4 Add Family Member Use Case

Use Case Attribute	Details
Use Case Name	Add Family Member
Primary Actors	Patient
Description	The patient can add family members to their account for shared health tracking.
Trigger	The patient wants to add a dependent.
Pre-Conditions	Patient must be logged in.
Post-Conditions	Family members are added successfully to the patient's account.
Normal Flow	1. Patient logs in → 2. Navigates to profile → 3. Adds family details → 4. Confirms addition
Alternative Flow	Invalid details provided
Exceptions	Server error, no internet, or duplicate entries
Business Rules	Each patient can add a limited number of dependents (configurable).

3.5.4 Book Appointments

The Book Appointments use case enables patients to schedule appointments within the Pulse Point system. This use case is triggered whenever a patient attempts to book a session with a doctor. The precondition requires that the doctor has available slots before an appointment can be scheduled.

The normal flow involves the user logging in, selecting a doctor and an available time slot from the displayed grid (30-minute slots between 9 AM and 5 PM, Monday through Saturday), and confirming the booking, which results in a successfully scheduled appointment stored in Firestore. Alternative flows handle cases where the chosen slot is unavailable (system suggests alternative slots), and exceptions include server errors or loss of internet connectivity. Business rules ensure that overbooking is strictly prohibited (double-booking prevented by atomic transactions), appointments cannot overlap, and cancellations require at least 2 hours notice.

Table 5 Book Appointments

Use Case Attribute	Details
Use Case Name	Book Appointments
Primary Actors	Patient
Description	Patients can schedule an appointment with a doctor.
Trigger	A patient wants to book a consultation.
Pre-Conditions	The doctor has available slots on the selected date.
Post-Conditions	The appointment is scheduled successfully in Firestore, notifications sent.
Normal Flow	1. Login → 2. Select doctor and time slot → 3. Confirm booking
Alternative Flow	Selected slot is unavailable → System suggests alternative slots
Exceptions	Server error, no internet connection
Business Rules	Overbooking is not allowed (atomic transactions prevent double-booking). Cancellations require at least 2 hours notice.

3.5.5 Book Appointments

The Accept/Decline Appointments use case allows doctors to manage appointment requests from patients within the Pulse Point system. This use case is triggered whenever a patient books an appointment. The pre-condition requires that the doctor has available slots to accommodate appointments. In the normal flow, the doctor logs in, navigates to the appointment requests, and either accepts or declines each request, with the patient being notified of the outcome via push notification. Alternative flows handle scenarios where appointments are automatically declined if the doctor does not respond within a specified time limit (planned for future version). Business rules mandate that doctors respond to appointment requests within a set timeframe to ensure proper scheduling and patient satisfaction.

Table 6 Accept/Decline Appointments

Use Case Attribute	Details
Use Case Name	Accept/Decline Appointments
Primary Actors	Doctor
Description	Doctors can accept or decline appointment requests from patients.
Trigger	A patient booked an appointment.
Pre-Conditions	The doctor has available slots.
Post-Conditions	The appointment is accepted or declined, and the patient is notified via push notification.
Normal Flow	1. Doctor logs in → 2. Navigates to appointment requests → 3. Accepts or declines
Alternative Flow	Appointment is auto-declined if not responded to within a time limit (future feature)
Exceptions	System downtime, doctor unavailability
Business Rules	Doctors should respond within a set time frame.

3.5.6 Access Patient Records

The Access Patient Records use case allows both patients and doctors to view health records within the Pulse Point system. This use case is triggered whenever a doctor needs to review a patient's medical history before a consultation or a patient wants to check their own data. The pre-condition requires that the patient has medical records stored in the system. Alternative flows handle cases where no records are available. Business rules ensure that only authorized users are allowed to access patient records: patients can access their own records, doctors can access records of patients they have appointments with, and admins can access all records for system management, maintaining privacy and data security.

Table 7 Access Patient Records Use Case

Use Case Attribute	Details
Use Case Name	Access Patient Records
Primary Actors	Patient, Doctor
Description	Both patients and doctors can access health records.
Trigger	A doctor needs patient history, or a patient checks their own data.
Pre-Conditions	Patient must have medical records stored in the system.
Post-Conditions	Requested data is successfully accessed and displayed.
Normal Flow	1. Login → 2. Navigate to records → 3. View history
Alternative Flow	No records available for this patient
Exceptions	Data retrieval error, no internet, or unauthorized access attempt
Business Rules	Only authorized users can access records. Patients access own records; doctors access records of patients they have appointments with.

3.5.7 Prescribe Medication

The Prescribe Medication use case enables doctors to issue prescriptions and patients to view them within the Pulse Point system. The pre-condition requires that the patient has a valid consultation (appointment status Scheduled or In Progress). In the normal flow, the doctor logs in, accesses the patient's data, enters medication details using a structured form (selecting medicine from admin-managed database, specifying dosage, duration, and special instructions), and issues the prescription, which is then saved in the system and linked to the patient's medical record

Table 8 Prescribe Medication Use Case

Use Case Attribute	Details
Use Case Name	Prescribe Medication
Primary Actors	Doctor
Description	The doctor can prescribe medication, and the patient can view prescriptions.
Trigger	A doctor needs to issue a prescription after consultation.
Pre-Conditions	Patient has a valid consultation (appointment exists).
Post-Conditions	Prescription is saved in the system and linked to patient's medical record.
Normal Flow	1. Doctor logs in → 2. Accesses patient data → 3. Enters medication details → 4. Issues prescription
Alternative Flow	Prescription approval required (future feature)
Exceptions	System error, no internet, or unauthorized access
Business Rules	Only licensed doctors can issue prescriptions. Only medicines in admin-managed database can be prescribed.

3.5.8 View Medical Records

The View Medical Records use case allows patients to access their past medical history, prescriptions, and test reports within the Pulse Point system. This use case is triggered whenever a patient wants to check their medical history. The pre-condition requires that the patient is logged in. In the normal flow, the patient logs in, navigates to the medical records section, and views the details organized into tabs for Past Appointments, Prescriptions, and Lab Results. Alternative flows account for situations where the patient has no records. Business rules ensure that only the patient and authorized doctors can view the records, maintaining privacy and security.

Table 9 View Medical Records Use Case

Use Case Attribute	Details
Use Case Name	View Medical Records
Primary Actors	Patient
Description	The patient can view their past medical history, prescriptions, and test reports.
Trigger	The patient wants to check their medical history.
Pre-Conditions	Patient must be logged in.
Post-Conditions	Patient successfully views medical records.
Normal Flow	1. Patient logs in → 2. Navigates to medical records → 3. Views details organized by tabs
Alternative Flow	Patient has no records
Exceptions	No internet connection, server down, or data retrieval error
Business Rules	Only the patient and authorized doctors can view records.

3.5.9 Request Ambulance

The Request Ambulance use case allows patients to request an emergency ambulance with real-time GPS tracking. This use case can be triggered even without logging in (emergency mode). The

system captures the patient's GPS location (or allows manual address entry), finds the nearest available ambulance by calculating distances using the Haversine formula, dispatches that unit, and provides a tracking screen showing the ambulance's real-time location on a map with estimated arrival time.

Table 10 Request Ambulance Use Case

Use Case Attribute	Details
Use Case Name	Request Ambulance
Primary Actors	Patient
Description	Patient can request an emergency ambulance with real-time GPS tracking.
Trigger	Patient taps "Request Ambulance" button.
Pre-Conditions	GPS enabled on device, at least one ambulance unit available.
Post-Conditions	Nearest ambulance dispatched, tracking screen displayed.
Normal Flow	1. Tap Request Ambulance → 2. Allow GPS permission → 3. Select ambulance type → 4. Tap Request Now → 5. System calculates nearest unit → 6. Dispatches ambulance → 7. Shows tracking screen with map and ETA
Alternative Flow	GPS not enabled → System prompts to enable GPS. No ambulances available → Error message displayed.
Exceptions	No internet connection, GPS unavailable, server error
Business Rules	Nearest ambulance assigned using Haversine formula. Emergency access does not require authentication. Only one ambulance dispatched per request.

3.5.10 Use AI Chatbot

The AI Chatbot use case allows patients to ask health-related questions and receive AI-generated responses. The system sends user queries to the Gemini API with a predefined prompt defining it as a medical assistant. The AI provides general health information, avoids diagnoses or prescriptions, and includes a disclaimer recommending consultation with a doctor. Responses are displayed in the chat interface.

Table 11 Use AI Chatbot Use Case

Use Case Attribute	Details
Use Case Name	Use AI Chatbot
Primary Actors	Patient
Description	Patient can ask health-related questions and receive AI-generated responses.
Trigger	Patient types a question in the chat input field.
Pre-Conditions	Patient is logged in (optional for basic access), internet connection available.
Post-Conditions	AI response displayed, conversation saved to Firestore.
Normal Flow	1. Type question → 2. Tap send → 3. System sends query to Gemini API → 4. API returns response → 5. Response displayed in chat bubble → 6. Conversation saved
Alternative Flow	API rate limit exceeded → Error message displayed. Content blocked by safety filters → Appropriate error message.
Exceptions	No internet connection, API key invalid, server error
Business Rules	AI does not provide medical diagnoses or prescriptions. Disclaimer displayed prominently. Conversation history stored per user.

3.5.11 Order Medicine

The AI Chatbot use case allows patients to ask health-related questions and receive AI-generated responses. The system sends user queries to the Gemini API with a predefined prompt defining it as a medical assistant. The AI provides general health information, avoids diagnoses or prescriptions, and includes a disclaimer recommending consultation with a doctor. Responses are displayed in the chat interface..

Table 12 Order Medicine Use Case

Use Case Attribute	Details
Use Case Name	Order Medicines
Primary Actors	Patient
Description	Patient can order medicines from the pharmacy with cart functionality and prescription upload.
Trigger	Patient adds medicines to cart and proceeds to checkout.
Pre-Conditions	Patient logged in, items in cart.
Post-Conditions	Order created, stock reduced, patient notified.
Normal Flow	1. Browse medicines → 2. Add to cart → 3. View cart → 4. Upload prescription (if required) → 5. Place order → 6. System checks stock → 7. Reduces inventory → 8. Creates order → 9. Shows confirmation
Alternative Flow	Medicine out of stock → Error message, item removed from cart. Prescription not uploaded for Rx medicine → Error message.
Exceptions	No internet connection, transaction failure, server error
Business Rules	Prescription-only medicines require uploaded prescription image. Stock checked atomically before order confirmation. Order status tracked through defined workflow.

3.5.12 Book Lab Test

The Book Lab Test use case allows patients to book laboratory tests. The patient views available tests with descriptions and fees, selects a test, views available time slots, chooses a date and time slot, confirms booking, waits for lab staff to upload results, receives a notification when results are available, and views/downloads the digital result from the medical records section.

Table 13 Book Lab Test Use Case

Use Case Attribute	Details
Use Case Name	Book Lab Test
Primary Actors	Patient
Description	Patient can book laboratory tests and receive digital results.
Trigger	Patient taps "Lab Tests" from dashboard.
Pre-Conditions	Patient logged in.
Post-Conditions	Lab booking created, notification sent when results available.
Normal Flow	1. View available tests → 2. Select test → 3. Choose date and time slot → 4. Confirm booking → 5. Receive confirmation → 6. Lab staff uploads results → 7. Notification received → 8. View/download result
Alternative Flow	No available slots → Message displayed. Results not yet uploaded → Status shows "Pending".
Exceptions	No internet connection, server error
Business Rules	Results uploaded by lab staff via admin panel. Patient notified when results available.

3.5.13: Approve Doctor Registration Use Case

The Approve Doctor Registration use case allows the admin to verify and approve doctor accounts within the Pulse Point system. This use case is triggered whenever a doctor submits a registration request. The pre-condition requires that the doctor has submitted valid documents (medical license, specialization certificates, clinic address). Normally, the admin logs in, reviews the request, verifies credentials, and approves or rejects it, activating or rejecting the doctor's account. The doctor receives a notification of the decision via email (Firebase) and push notification (FCM). Business rules ensure that only verified doctors are approved, maintaining system integrity and quality of care.

Table 14 Approve Doctor Registration Use Case

Use Case Attribute	Details
Use Case Name	Approve Doctor Registration
Primary Actors	Admin
Description	Admin verifies and approves doctor accounts.
Trigger	A doctor submits a registration request.
Pre-Conditions	Doctor has submitted valid documents (medical license, certificates).
Post-Conditions	Doctor's account is activated or rejected, notification sent.
Normal Flow	1. Admin logs in → 2. Navigates to pending verifications → 3. Reviews doctor credentials → 4. Approves or rejects → 5. System updates doctor's verified status → 6. Notification sent to doctor
Alternative Flow	Approval is delayed for further verification (admin can mark as "under review")
Exceptions	Server issues, invalid documents uploaded
Business Rules	Only verified doctors can log in and access doctor features. Admins cannot approve themselves.

3.5.14 Manage Users

The Manage Users use case enables the admin to add, suspend, or remove users within the Pulse Point system. This use case is triggered when user activity is monitored or when a user reports an issue. The pre-condition requires that the user exists in the system. Normally, the admin logs in, navigates to user management, searches for the user, and edits the user's status (active/suspended/deleted). Business rules enforce that only authorized admins with proper permissions can modify user accounts, ensuring system security.

Table 15 Manage Users Use Case

Use Case Attribute	Details
Use Case Name	Manage Users
Primary Actors	Admin
Description	Admin can add, suspend, or remove users.
Trigger	User activity monitoring or issue reported.
Pre-Conditions	User exists in the system.
Post-Conditions	User status is updated (active/suspended/deleted).
Normal Flow	1. Login → 2. Navigate to user management → 3. Search for user → 4. Edit user status → 5. Save changes
Alternative Flow	Unauthorized admin tries to modify data → Access denied.
Exceptions	Server error, no internet connection
Business Rules	Only authorized admins with proper permissions can modify user accounts.

3.5.15 Manage Medicines

The Manage Medicines use case enables the admin to add, update, or delete medicines in the inventory database. The admin can add new medicines with name, category, price, initial stock, prescription requirement flag, and description; update existing medicines (change price, update stock levels, modify description); and delete medicines that are discontinued. Business rules ensure that medicine stock levels are accurate and that only admins can modify the medicine database.

Table 16 Manage Medicines Use Case

Use Case Attribute	Details
Use Case Name	Manage Users
Primary Actors	Admin
Description	Admin can add, suspend, or remove users.
Trigger	User activity monitoring or issue reported.
Pre-Conditions	User exists in the system.
Post-Conditions	User status is updated (active/suspended/deleted).
Normal Flow	1. Login → 2. Navigate to user management → 3. Search for user → 4. Edit user status → 5. Save changes
Alternative Flow	Unauthorized admin tries to modify data → Access denied.
Exceptions	Server error, no internet connection
Business Rules	Only authorized admins with proper permissions can modify user accounts.

3.5.16 Monitor System Logs

The Monitor System Logs use case allows admins to track system activity and review security logs within the Pulse Point system. This use case is triggered whenever an issue is reported or for periodic security audits. The pre-condition requires that system logs are available. Normally, the admin logs in, navigates to system logs, and reviews reports of user logins, appointment bookings, order placements, ambulance requests, and error events. Business rules ensure that all logs are stored securely and retained for a defined period.

Table 17 Monitor System Logs Use Case

Use Case Attribute	Details
Use Case Name	Monitor System Logs
Primary Actors	Admin
Description	Admins can track system activity and security logs.
Trigger	An issue is reported or periodic security audit.
Pre-Conditions	System logs are available in Firestore.
Post-Conditions	System logs are reviewed, anomalies detected if any.
Normal Flow	1. Login → 2. Navigate to system logs → 3. Filter by date, user, or event type → 4. Review reports
Alternative Flow	No logs available for selected filter → Empty state message.
Exceptions	Corrupt logs, server error
Business Rules	Logs must be stored securely and retained for defined period (e.g., 90 days).

3.6 Functional Requirements

Functional requirements define the specific behaviors and operations that the Pulse Point system must provide to satisfy the needs of its users. These requirements outline the essential capabilities that the system must possess to manage patient health data, appointments, emergency ambulance dispatch, AI-powered medical assistance, pharmacy ordering, laboratory test booking, and administrative tasks effectively. The Pulse Point system is designed to offer a secure, reliable, and user-friendly environment that ensures data integrity, privacy, and efficient healthcare service management.

3.6.1 User Management

The Pulse Point system allows patients, doctors, and administrators to register using valid credentials such as email, phone number, and secure passwords. Registration is guided to ensure all required fields are completed correctly, and login is secured with Firebase Authentication and role-based access. Patients access personal records and appointments, doctors access patient data and consultation tools, and administrators manage system functions. Security measures include hashed passwords (bcrypt via Firebase), email verification for new accounts, and optional rate limiting for failed login attempts. Repeated failed logins may trigger temporary account locks.

Users can manage and update their profiles after registration. Patients can maintain contact details, date of birth, blood group, and emergency contacts. Doctors provide professional details such as specialization, experience years, clinic address, consultation fee, and availability hours, which can be updated dynamically. The system validates all changes and sends automated notifications to keep users informed of profile updates or verification needs.

Account verification ensures authenticity. Administrators review doctor registrations, validate documents, and approve or reject accounts, while the system notifies users of their registration status. Patients may verify accounts via email. All verification activities are logged to maintain transparency and accountability.

3.6.2 Patient Records Management

The system provides secure access to patient health records, enabling patients to view their complete medical history, including consultations, prescriptions, test results, and treatment plans. A clear, user-friendly interface helps patients monitor their health and make informed decisions.

Doctors can access patient records during consultations to provide accurate diagnoses and treatment. Strict access control ensures only authorized users can view or modify records, and all access and modifications are logged for accountability and compliance. Data security is critical; all records are encrypted and protected against unauthorized access via Firestore Security Rules.

3.6.3 Appointments Management

The system allows patients, doctors, and administrators to schedule and manage appointments efficiently. Patients can book appointments using an interface that displays available doctor slots.

(30-minute slots between 9 AM and 5 PM, Monday through Saturday), preventing overlaps and ensuring optimal resource use. The system uses atomic transactions to prevent double-booking.

Doctors can accept or decline requests, with immediate push notifications sent to patients. Administrators can modify, reschedule, or cancel appointments while enforcing access control to prevent unauthorized changes. Automated reminders and alerts keep users informed of upcoming appointments or changes, improving coordination, adherence to schedules, and overall service efficiency.

3.6.4 Ambulance Dispatch Management

The system allows patients to request emergency ambulances with real-time GPS tracking. The patient's location is captured via device GPS, and the system queries Firestore for available ambulance units. For each available unit, the system calculates the distance to the patient using the Haversine formula (great-circle distance on Earth). The unit with the minimum distance is selected and dispatched, with its status updated to "Dispatched". The patient receives a tracking screen with a map showing the ambulance's real-time location and estimated arrival time. Push notifications are sent to the ambulance driver and the hospital emergency department. This use case can be triggered even without logging in (emergency mode).

3.6.5 AI Chatbot

The system provides an AI-powered medical chatbot using the Gemini API. Patients can ask health-related questions in natural language and receive instant responses. The system sends the query to the Gemini API with a system prompt that defines the AI as a medical assistant with the following guidelines: provide basic health information only (symptoms, general advice), do NOT provide medical diagnoses, do NOT prescribe medications, for emergencies always say "Seek immediate medical attention", always include a disclaimer about consulting a doctor, and be helpful, concise, and professional. The API returns a response which is displayed in the chat interface. A prominent disclaimer is displayed at the top of the chat screen reminding users that the AI is not a substitute for professional medical advice.

3.6.6 Pharmacy Ordering and Inventory Management

The system allows patients to browse medicines by category, add them to a cart, upload prescription images for prescription-only medicines, place orders, and track order status (Pending → Confirmed → Processing → Out for Delivery → Delivered). Pharmacy staff (via admin panel) can update order status. The system checks stock availability before order confirmation using atomic Firestore transactions, reducing inventory count atomically to prevent overselling when multiple customers order the same last unit simultaneously. Business rules ensure that prescription-only medicines cannot be ordered without an uploaded prescription image, stock is checked before order confirmation, and order status updates trigger push notifications to the patient.

3.6.7 Laboratory Test Booking and Results

The system allows patients to book laboratory tests, select date and time slots from available options, receive confirmation notifications, and later view digital results when uploaded by lab staff. Lab staff (via admin panel) can upload result files (PDF or image) to Cloud Storage and link them to the patient's booking record. Patients receive a push notification when results are available and can view or download the results from the medical records section.

3.6.8 Prescriptions and Medications

Doctors can issue prescriptions securely after consultations, and prescriptions are linked to patient records. Each prescription contains detailed information about medication (selected from admin-managed medicine database), dosage, duration, and special instructions. The system ensures that only licensed doctors can issue prescriptions, preventing misuse. Patients can access and review prescriptions to follow their treatment accurately. Prescription data is securely stored in Firestore and linked to both the patient and the appointment, and all prescriptions are logged for compliance and safety.

3.6.9 Notifications and Alerts

The system provides a comprehensive notification mechanism to keep users informed of important updates. Patients and doctors receive reminders for upcoming appointments via push notifications (Firebase Cloud Messaging), reducing missed consultations and ensuring timely healthcare delivery. Any changes to appointments automatically trigger alerts for both parties to maintain transparency and prevent scheduling conflicts. Notifications also cover order status updates (order

placed, confirmed, out for delivery, delivered), ambulance dispatch alerts (request confirmed, unit dispatched, unit arrived), lab result availability, profile updates, prescription approvals, and account modifications.

3.6.10 Admin and System Management

Administrators have comprehensive tools to manage users, monitor system activity, and enforce security policies. They can view all registered users (patients and doctors), add new users, suspend or delete user accounts, assign roles, and reset passwords. Doctor verification involves reviewing uploaded credentials (medical license, specialization certificates) and approving or rejecting applications. Medicine management includes adding new medicines with name, category, price, stock, prescription requirement flag; updating existing medicines; removing discontinued medicines; and managing stock levels.

System monitoring includes viewing user activity logs (logins, appointment bookings, order placements, ambulance requests), error logs for debugging, and generating reports on appointments by doctor and date, orders by medicine and status, and user registrations over time. All actions and critical events are logged for auditing and accountability. Role-based access control ensures that users can only access features and data relevant to their role, protecting sensitive information while allowing efficient task execution.

3.6.11 Data Reporting and Analytics

The Pulse Point system provides reporting and analytics to support informed decision-making. Administrators can generate financial reports to track revenue streams and resource allocation. Analytics on system usage, appointment trends, order trends, ambulance response times, and user engagement help optimize performance and improve user experience. All reporting and analytics features rely on secure data handling and real-time processing to ensure accurate, up-to-date insights. Automated report generation and alerts further support proactive management of healthcare services and administrative operations.

3.7 Non-Functional Requirements

Non-functional requirements define the quality attributes, performance standards, and operational constraints that the Pulse Point system must meet. While functional requirements describe what

the system does, non-functional requirements specify how the system performs and the standards it must uphold in terms of reliability, security, usability, and efficiency. These requirements are critical for ensuring that the Pulse Point system operates effectively under real-world conditions, meets user expectations, and complies with healthcare regulations and standards.

3.7.1 Performance

The Pulse Point system must be capable of handling multiple simultaneous users without noticeable delays, ensuring fast response times for all operations, including accessing medical records, booking appointments, requesting ambulances, ordering medicines, using the AI chatbot, and processing payments (when implemented). The system should be optimized for high performance so that queries on patient data, prescription history, appointment availability, and financial transactions are executed efficiently. Adequate server resources (Firebase's scalable infrastructure), database optimization (indexes on frequently queried fields), and efficient algorithms (Haversine distance calculation, $O(n)$ nearest ambulance search) must be implemented to maintain consistent performance even during peak usage hours.

Specific performance targets: App cold start ≤ 3 seconds, screen transitions ≤ 0.5 seconds, Firestore queries ≤ 200 ms (95th percentile), AI chatbot response ≤ 5 seconds (95th percentile), nearest ambulance calculation (10 units) ≤ 1 second.

3.7.2 Scalability

The system must be scalable to accommodate growth in the number of users, patient records, appointments, ambulance requests, pharmacy orders, and medical data over time. As more patients, doctors, and administrators use the platform, the system should be able to scale horizontally or vertically without affecting performance. Firebase Firestore automatically scales to handle large datasets and high concurrency. This ensures that Pulse Point can expand its services to larger healthcare networks or multiple clinics while maintaining responsiveness and stability.

3.7.3 Security

Security is a critical non-functional requirement for Pulse Point due to the sensitivity of patient health information, medical records, and personal data. The system must implement encryption for data at rest (Firestore and Cloud Storage encryption, AES-256) and in transit (HTTPS/TLS 1.2+),

ensuring that all patient records, prescriptions, payment details (when implemented), and personal information are protected against unauthorized access. Role-based access control must be strictly enforced via Firestore Security Rules to ensure that only authorized users can view or modify sensitive information. Additionally, the system must support secure authentication methods, such as strong passwords (minimum 8 characters), email verification, and rate limiting for failed login attempts. Regular security audits and monitoring must be conducted to detect and mitigate potential vulnerabilities.

3.7.4 Reliability and Availability

Pulse Point must provide high reliability and ensure continuous availability for all users, including patients, doctors, and administrators. The system should have minimal downtime, and robust error-handling mechanisms must be in place to manage server failures, network disruptions, or unexpected exceptions. Firebase provides high availability (99.95% uptime SLA for Firestore) with automatic failover and redundancy. Real-time monitoring of system health is essential to maintain 24/7 availability. Automatic backup procedures (Firestore automated daily backups) and recovery mechanisms must be implemented to ensure uninterrupted access and protection of critical data.

3.7.5 Usability

The system must offer an intuitive and user-friendly interface suitable for users with varying levels of technical expertise. Patients, doctors, and administrators should be able to navigate the system easily to perform tasks such as viewing medical records, booking appointments, requesting ambulances, ordering medicines, issuing prescriptions, or managing users. Clear navigation, consistent design (following Material Design guidelines), accessible forms with validation, large touch targets (minimum 48x48 dp for buttons), appropriate color contrast (WCAG AA standards), and contextual help features are required to enhance the user experience and reduce errors. The System Usability Scale target is 4.5 out of 5 based on testing with healthcare professionals.

3.7.6 Maintainability

Pulse Point must be designed for ease of maintenance and updates. The system should support modular architecture (separation into feature modules: auth, appointment, ambulance, pharmacy, lab, chatbot, records, admin), allowing components to be updated or replaced without affecting

other functionalities. Clear documentation (code comments for complex logic, separate SRS and SDD documents), coding standards (Flutter and Dart best practices), and version control practices (Git with meaningful commit messages, feature branches) must be followed to facilitate efficient maintenance, debugging, and future enhancements.

3.7.7 Compliance

The system must comply with healthcare regulations, data protection laws, and industry standards to ensure that patient information is handled legally and ethically. All operations, from storing medical records to processing orders and ambulance requests, must adhere to these compliance standards to protect patients and healthcare providers from legal and operational risks. The system is designed with privacy by design principles, including data minimization (only necessary data collected), purpose limitation (data used only for intended purposes), and access controls.

3.7.8 Backup and Recovery

The Pulse Point system must implement regular data backups and provide reliable recovery mechanisms. Firebase Firestore automatically performs daily backups with point-in-time recovery. Backups are stored securely, ensuring that patient records, prescriptions, financial data (when implemented), appointment details, ambulance requests, and pharmacy orders can be restored in the event of system failure, accidental deletion, or data corruption. Recovery procedures must be tested periodically to verify effectiveness and minimize downtime during emergencies.

3.7.9 Performance Monitoring and Logging

The system must include mechanisms for monitoring performance, logging user activity, and detecting anomalies. Firebase provides monitoring through the Firebase Console with metrics for Firestore reads/writes, Authentication usage, and Cloud Functions execution. Custom logging can be implemented for user actions (logins, bookings, orders). Real-time monitoring of server load, database performance, and application errors ensures that potential issues are identified and addressed proactively.

Chapter 4

DESIGN AND ARCHITECTURE

4. Design and Architecture

The Design and Architecture chapter of the Pulse Point system explains how the software is structured, how its components interact, and how data flows throughout the system. This chapter provides a detailed overview of the system's architecture, data representation, process flows, and design models. It highlights the technical decisions made to ensure performance, security, scalability, and maintainability. By presenting the system design clearly, this chapter demonstrates how Pulse Point meets both functional and non-functional requirements while delivering a reliable healthcare management platform.

4.1 System Architecture

The system architecture defines the overall structure of Pulse Point and illustrates how its main components work together to deliver healthcare services efficiently. It ensures that each module performs its specific tasks while interacting seamlessly with others to maintain smooth operations and data integrity.

4.1.1 Architecture Style

Pulse Point is developed using a layered client-server architecture. The client layer consists of the Flutter-based mobile interface that allows patients, doctors, and administrators to interact with the system. The server layer includes Firebase backend services (Authentication, Firestore, Cloud Messaging, Cloud Storage) that manage authentication, data storage, push notifications, and file storage. External services such as the Gemini API and Google Maps API integrate via REST APIs. This separation of concerns enhances maintainability, supports scalability as the number of users grows, and ensures secure, organized handling of system operations.

4.1.2 Components

The Pulse Point system is composed of several interconnected components, each designed to perform a specific function within the overall architecture. These components work collaboratively to ensure smooth system operation, secure data handling, and efficient healthcare service delivery. By dividing the system into modular parts, Pulse Point maintains clarity in functionality, supports easier maintenance, and allows individual components to evolve without affecting the entire system.

4.1.2.1 User Interface (UI)

The UI provides mobile access to all user types: patients, doctors, and administrators. It is designed to be intuitive and responsive, ensuring users can navigate features such as registration, profile management, appointment scheduling, ambulance requests, pharmacy ordering, lab test booking, AI chatbot, prescription viewing, medical record access, and financial transactions (when implemented) with ease. The UI serves as the primary point of interaction between the user and the system.

4.1.2.2 Authentication Service

The authentication service manages user registration, login, and role-based access control. It ensures that only authorized users can access the system according to their roles, implementing security measures such as hashed password storage (bcrypt via Firebase), email verification, and monitoring of failed login attempts (rate limiting). This component safeguards sensitive data and prevents unauthorized access.

4.1.2.3 Patient Records Module

This module stores and manages comprehensive patient health information, including medical histories, prescriptions, diagnostic results, and treatment plans. It implements strict access control to ensure that only patients and their authorized doctors can view or modify records. Firestore Security Rules and audit logs track all access and changes, maintaining transparency and compliance with healthcare regulations.

4.1.2.4 Appointments Module

The appointments module facilitates booking, rescheduling, and managing appointments. It displays available time slots based on doctor schedules (30-minute slots, 9 AM to 5 PM, Monday through Saturday) and ensures no conflicts occur by using atomic Firestore transactions. Doctors can accept or decline requests, while push notifications keep patients informed. Administrators can modify schedules when necessary, ensuring efficient allocation of resources and smooth workflow management.

4.1.2.5 Ambulance Dispatch Module

The ambulance dispatch module handles emergency ambulance requests. It captures the patient's GPS location, queries Firestore for available ambulance units, calculates distances using the Haversine formula, selects the nearest unit, dispatches it, and provides real-time tracking on a map via Google Maps API. The module sends push notifications to the ambulance driver and the hospital emergency department.

4.1.2.6 Pharmacy Module

The pharmacy module manages medicine inventory and order processing. It allows patients to browse medicines by category, view details with price and stock status, add to cart, upload prescription images for prescription-only medicines, place orders, and track order status. Pharmacy staff (via admin panel) can update order status, and the module automatically updates inventory counts when orders are confirmed using atomic transactions.

4.1.2.7 Laboratory Module

The laboratory module allows patients to book lab tests, select available time slots, and receive digital results when uploaded. Lab staff (via admin panel) can upload result files (PDF or image) to Cloud Storage and link them to the patient's booking record. Patients receive push notifications when results are available.

4.1.2.8 AI Chatbot Module

The AI chatbot module integrates with the Gemini API to provide AI-powered medical assistance. It sends user queries to the API with a system prompt defining the AI as a medical assistant, receives responses, and displays them in the chat interface with appropriate disclaimers. Conversation history is stored in Firestore for context in continuing conversations.

4.1.2.9 Admin Dashboard

The admin dashboard provides tools for administrators to manage users, verify doctors, manage medicines, monitor system activity (logs), and generate reports. It allows oversight of appointments, orders, ambulance requests, and system health, enabling effective management, security enforcement, and operational decision-making.

4.1.3 Interactions

All components communicate primarily through Firebase SDKs and RESTful APIs, ensuring efficient and standardized data exchange. The UI sends requests to Firebase services (Authentication, Firestore, Cloud Messaging, Cloud Storage), which process data and respond with relevant information. For instance, when a patient books an appointment, the UI sends data to Firestore, which triggers a Firestore listener or Cloud Function to send a push notification to the doctor via FCM. When a patient requests an ambulance, the UI captures GPS location, sends the request to Firestore, and the nearest ambulance calculation runs on the client side using Firestore data. When a patient asks a question to the AI chatbot, the UI sends an HTTP request directly to the Gemini API (or via a Cloud Function for added security). All communication is secured using HTTPS protocols and authentication tokens to protect data integrity and user privacy.

4.1.4 System Architecture Diagram Description

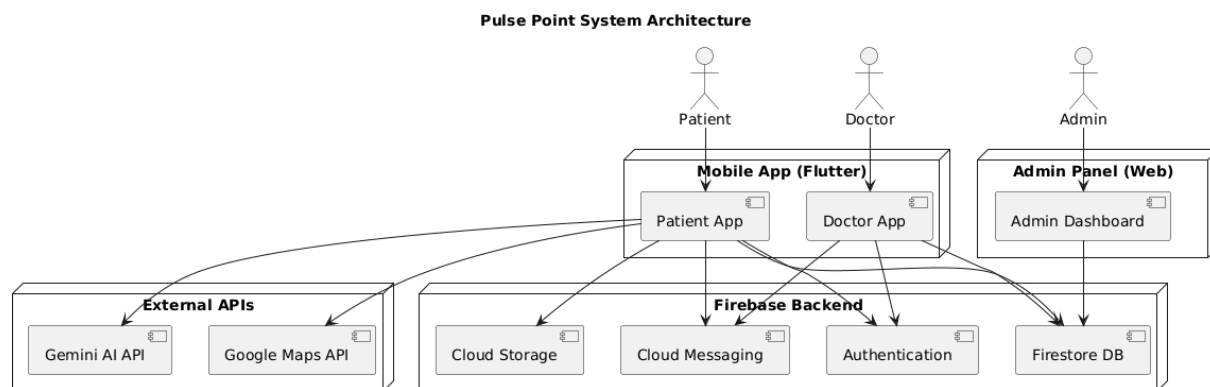


Figure 4.1 System Architecture

The system architecture diagram illustrates the overall structure and interaction of the Pulse Point system components. It shows how users including patients, doctors, and administrators interact with the application through the mobile interface. The diagram highlights the flow of information between the user interface, Firebase backend services (Authentication, Firestore, Cloud Messaging, Cloud Storage), and external APIs (Gemini API, Google Maps API).

The architecture is designed to ensure smooth communication between modules and efficient data management. When a user performs an action, such as logging in, registering, booking an appointment, requesting an ambulance, ordering a medicine, or asking the AI chatbot, the request is sent to the appropriate backend service. The backend processes the request, interacts with the

database or external API to retrieve or store data, and then returns the response to the user interface. This structure provides a clear separation of concerns, improves system scalability, and enhances maintainability.

4.2 Data Representation

The Data Representation section explains how information is structured, stored, and managed within the Pulse Point system. It covers the database design approach, entity relationships, data flow between modules, and the organization and security of stored data. This ensures that all system components can efficiently access and process the information they need while maintaining accuracy, consistency, and privacy.

4.2.1 Database Design

Pulse Point uses Firebase Firestore, a NoSQL cloud database, due to its flexibility, scalability, real-time capabilities, and ability to handle large volumes of semi-structured medical data. Since health records vary greatly in format such as prescriptions, test results, medical histories, appointment logs, ambulance requests, and pharmacy orders, Firestore allows storing this information in dynamic document structures. This design supports rapid read/write operations, horizontal scalability, real-time listeners for automatic UI updates, and easy integration with cloud-based healthcare solutions.

The database is organized into the following collections:

users: Stores basic user information common to all user types. Fields: `userId` (document ID, matches Firebase Auth UID), `name`, `email`, `phone`, `role` (patient/doctor/admin), `profileImageUrl` (optional), `createdAt` timestamp, `updatedAt` timestamp.

patients: Stores patient-specific information, extending the users collection. Fields: `patientId` (document ID, matches `userId`), `dateOfBirth`, `bloodGroup`, `emergencyContact` (name and phone), `familyMemberIds` (array of user IDs for dependents).

doctors: Stores doctor-specific information, extending the users collection. Fields: `doctorId` (document ID, matches `userId`), `specialization`, `experienceYears`, `clinicAddress`, `consultationFee` (number), `availableDays` (array of integers 0-6), `availableHoursStart` (string, e.g., "09:00"), `availableHoursEnd` (string, e.g., "17:00"), `verified` (boolean), `medicalLicenseUrl` (optional).

appointments: Stores appointment records. Fields: appointmentId (document ID, auto-generated), patientId (reference to users), doctorId (reference to users), date (timestamp), time (string, e.g., "10:30"), status (Scheduled/Completed/Cancelled/NoShow), patientNotes (optional), doctorNotes (optional), prescriptionId (optional reference), createdAt timestamp, updatedAt timestamp.

ambulance_requests: Stores ambulance request records. Fields: requestId (document ID), patientId (optional, null for emergency mode), pickupLat (number), pickupLng (number), pickupAddress (string, from reverse geocoding), ambulanceType (Basic/Advanced/ICU), assignedUnitId (reference to ambulance_units), status (Pending/Dispatched/Arrived/Completed), createdAt timestamp.

medicines: Stores pharmacy inventory. Fields: medicineId (document ID), name, category (Pain Relief/Antibiotics/Vitamins/Chronic Disease/Other), price (number), stock (integer), prescriptionRequired (boolean), description (optional), imageUrl (optional).

pharmacy_orders: Stores pharmacy orders. Fields: orderId (document ID), patientId (reference to users), items (array of objects: {medicineId, name, quantity, price}), totalAmount (number), status (Pending/Confirmed/Processing/OutForDelivery/Delivered), prescriptionImageUrl (optional), createdAt timestamp, updatedAt timestamp.

lab_tests: Stores available laboratory tests. Fields: testId (document ID), name, description, fee (number), preparationInstructions (optional).

lab_bookings: Stores laboratory test bookings. Fields: bookingId (document ID), patientId (reference to users), testId (reference to lab_tests), date (timestamp), timeSlot (string), status (Scheduled/Completed/Cancelled), resultUrl (optional), resultUploadedAt timestamp (optional).

chat_sessions: Stores AI chatbot conversation history. Fields: sessionId (document ID), userId (reference to users), createdAt timestamp, lastUpdatedAt timestamp, clearedAt timestamp (optional). Subcollection: messages with fields role (user/assistant), content (text), timestamp.

4.2.2 Entity-Relationship Representation

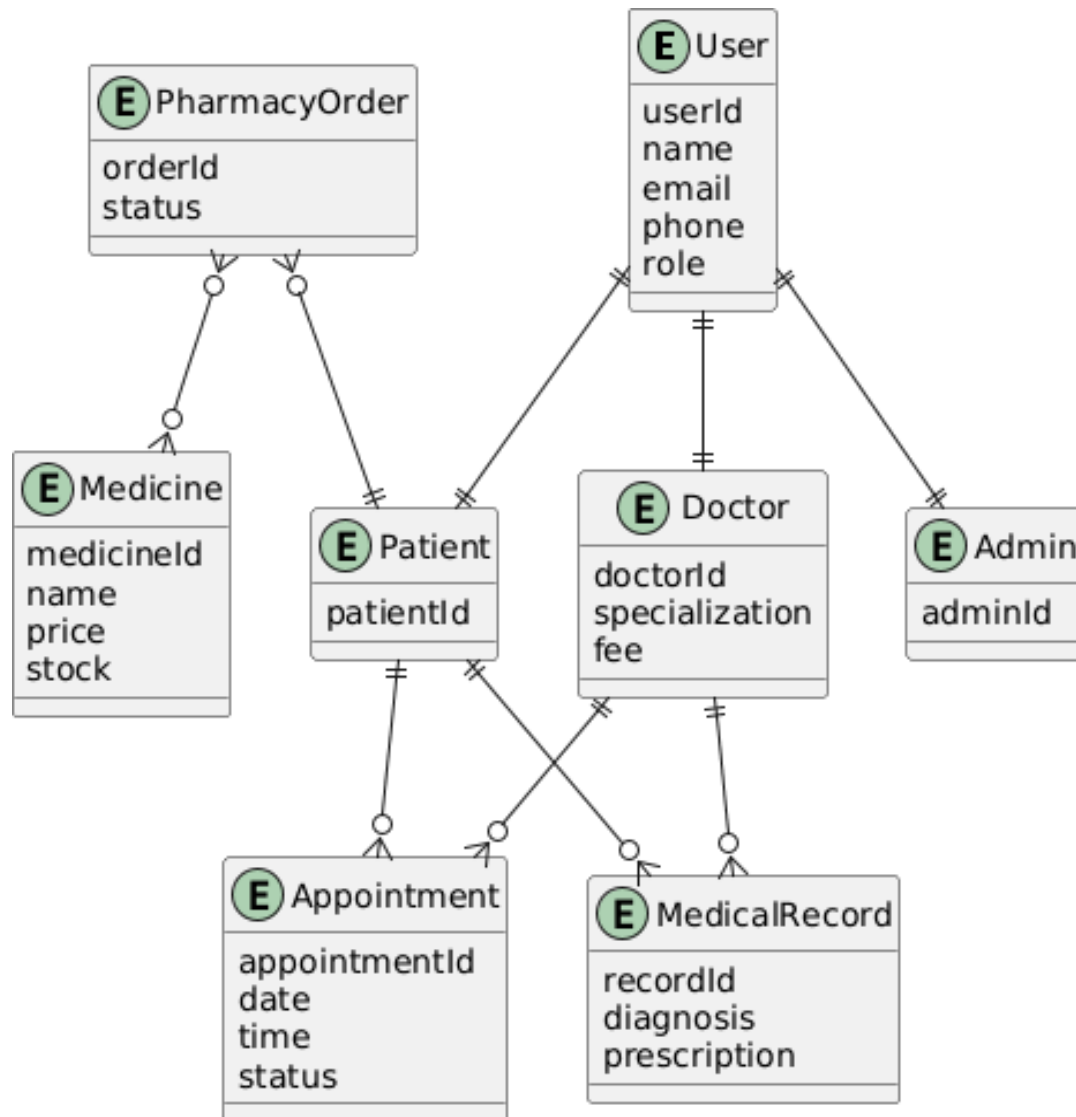


Figure 4.2 ER Diagram

Although Firestore does not use traditional relational tables, the system follows a structured entity model to maintain consistency. Each entity is stored as a document with well-defined attributes. Relationships are implemented using document references or embedded documents depending on access frequency and data size.

User Hierarchy: Patient, Doctor, and Admin entities are stored in separate collections but reference the same `userId` from the users collection. This allows common authentication data to be stored once while role-specific data is stored separately.

Appointment Relationships: Each appointment references a `patientId` and `doctorId`. This allows queries to fetch all appointments for a given patient (patient's appointment history) or for a given doctor (doctor's schedule). The relationship is one-to-many: one patient can have many appointments, one doctor can have many appointments.

Ambulance Request Relationships: Each ambulance request references a `patientId` (if logged in) and an `assignedUnitId`. This allows tracking of which patient requested which ambulance and which unit was dispatched.

Pharmacy Order Relationships: Each order references a `patientId`. The `items` field contains an array of objects, each referencing a `medicineId` (but also storing name and price denormalized for historical accuracy). This allows the order to remain accurate even if the medicine catalog changes later.

Lab Booking Relationships: Each lab booking references a `patientId` and a `testId`. When results are uploaded, the `resultUrl` field stores the Cloud Storage URL, and the `resultUploadedAt` timestamp records when results became available.

4.2.3 Data Flow

Data in the Pulse Point system moves through a well-defined pathway starting from the user interface and ending in secure storage. When a patient books an appointment, the information is captured in the presentation layer (Flutter UI), validated on the client side, and then saved to the Firestore database via the Firebase SDK. Doctors later retrieve this stored data through their dashboards for consultations. Similarly, ambulance request information flows from the UI (with GPS location) to Firestore, where the nearest ambulance calculation runs on the client side using fetched data. Pharmacy order-related information flows from the UI (cart items, prescription images uploaded to Cloud Storage) to Firestore, where inventory is updated atomically within a transaction.

4.2.3.1 Sequence Diagram Description

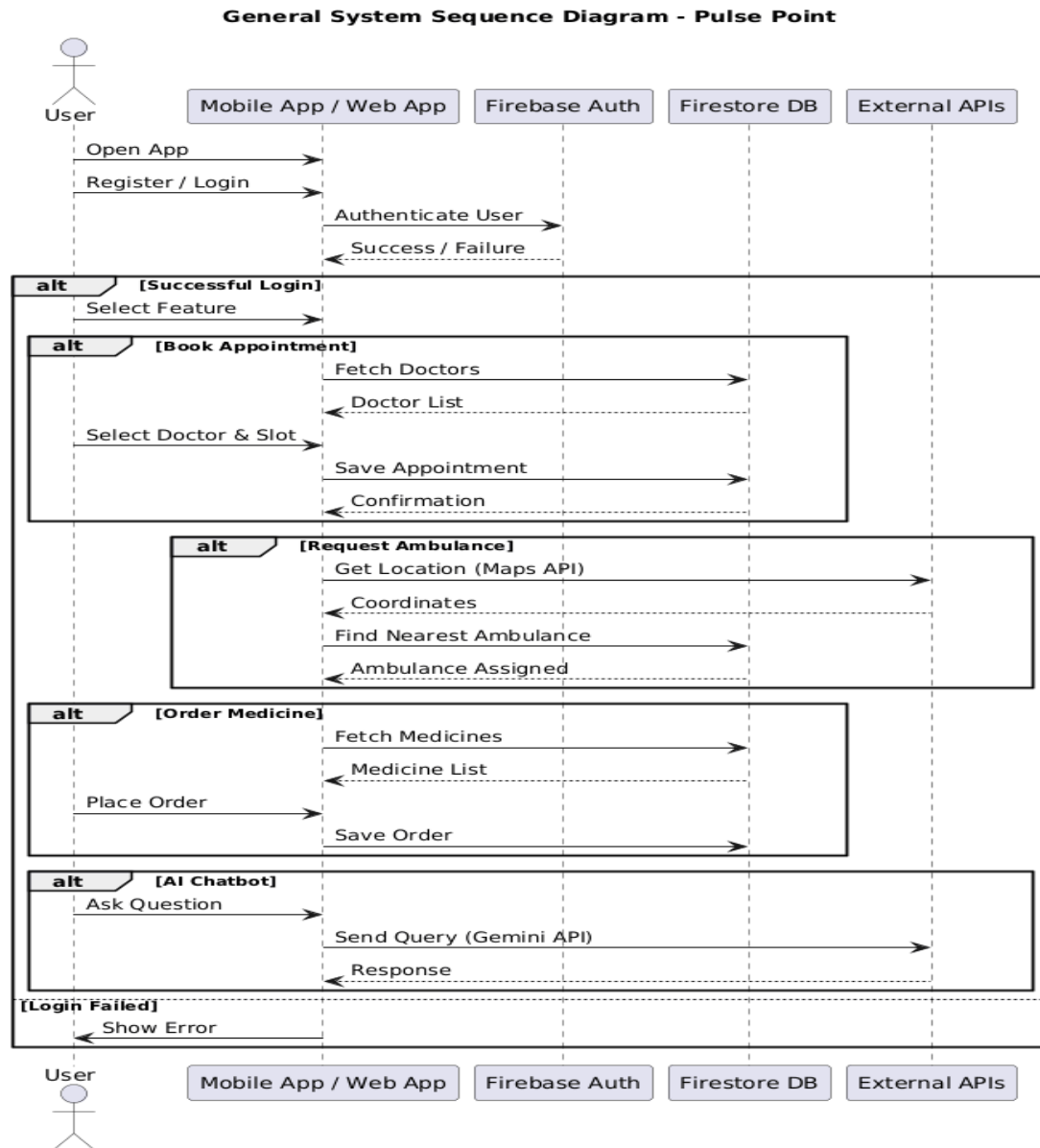


Figure 4.3 Sequence Diagram

The sequence diagram illustrates the communication between system components during important operations. For example, during the appointment scheduling process, the sequence begins with the patient (actor) interacting with the UI (Flutter widget), which sends a request to the AppointmentService. The AppointmentService calls Firestore to query available slots, Firestore returns the data, the UI displays slots, the patient selects a slot, the AppointmentService uses a Firestore transaction to check availability and create the appointment, Firestore confirms

success, the AppointmentService calls NotificationService to send push notifications via FCM, and notifications are delivered to the patient and doctor devices. This sequence highlights the order of interactions and clarifies how messages flow between components during each step of the process.

4.2.3.2 Activity Diagram Description

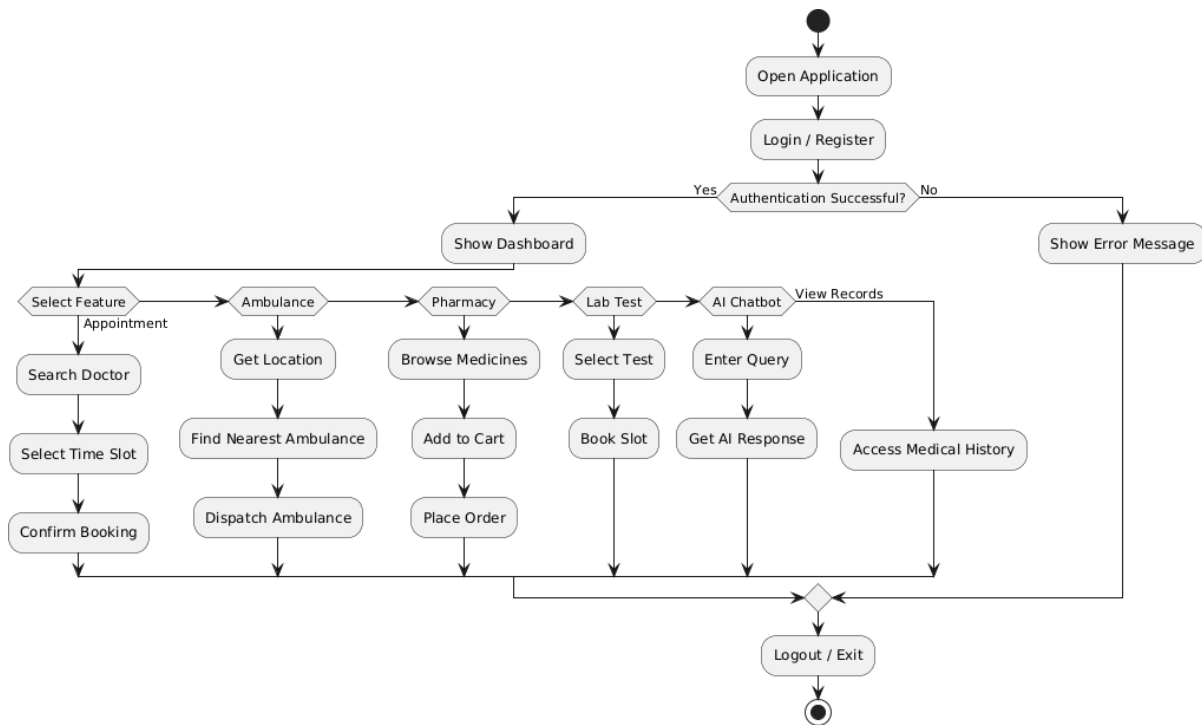


Figure 4.4 Activity Diagram

The activity diagram explains the workflows within the system by describing the actions, decisions, and outcomes that occur during various processes. In the ambulance dispatch activity, the user requests an ambulance, the system checks GPS permission (if not granted, requests permission; if denied, shows error), captures location, queries available ambulances, calculates nearest unit, dispatches unit, provides tracking, and ends. Decision nodes include: permission granted? yes/no; ambulances available? yes/no; dispatch successful? yes/no. This representation demonstrates how the system handles different scenarios and ensures that all necessary steps are followed according to predefined rules and logic.

4.2.4 Entity Descriptions and Constraints

Each entity within the database serves a specific purpose with constraints to ensure data integrity.

User: Unique `userId` (from Firebase Auth), email must be unique, role must be one of "patient", "doctor", or "admin". Required fields: `userId`, `name`, `email`, `role`, `createdAt`.

Patient: `patientId` must match an existing `userId` in `users` collection with role "patient". `dateOfBirth` must be a valid date in the past. `emergencyContact` should be a string in format "name: phone".

Doctor: `doctorId` must match an existing `userId` in `users` collection with role "doctor". `verified` defaults to false, can be updated only by admin. `consultationFee` must be a positive number. `availableDays` array values must be 0-6.

Appointment: `patientId` and `doctorId` must reference existing users. `date` must be a valid future date (or past for history). `time` must be in 30-minute increments between 09:00 and 17:00. `status` must be one of the enumerated values. Cancellations require at least 2 hours notice (enforced in application logic).

Ambulance Unit: `status` must be one of Available, Dispatched, Busy, Offline. `currentLat` and `currentLng` must be valid GPS coordinates. `lastLocationUpdate` should be updated whenever location changes.

Pharmacy Order: status transitions follow defined workflow (Pending → Confirmed → Processing → OutForDelivery → Delivered). `totalAmount` must equal sum of (`quantity` × `price`) for all items. For `prescriptionRequired` medicines, `prescriptionImageUrl` must not be null.

Security Constraints: Firestore Security Rules enforce role-based access: patients can read/write only their own data (appointments where `patientId` matches, orders where `patientId` matches, lab bookings where `patientId` matches); doctors can read appointments where `doctorId` matches and patient records for those appointments; admins can read/write all collections for system management. All data is encrypted at rest (Firestore default) and in transit (HTTPS).

4.3 Process Flow / Representation

The Process Flow section explains how key operations are carried out in the Pulse Point system. Each workflow is described clearly through structured explanations. These flows help readers understand the triggers, steps, and outcomes involved in major system processes.

4.3.1 Login and Authentication Process

The login and authentication process begins when a user enters their credentials through the Pulse Point interface. The system receives the input and verifies it by comparing the credentials with the records stored in Firebase Authentication. If the information matches, the system queries Firestore to retrieve the user's role (patient, doctor, or admin) from the users collection. For doctor accounts, the system also checks if the account has been verified by an admin. After successful verification, a secure session token is generated and managed by Firebase Authentication, allowing the user to access the system without repeatedly logging in. If the credentials are incorrect, the system rejects the login attempt, records the failed attempt, and prompts the user to try again, ensuring secure access control.

4.3.2 Appointment Booking Process

The appointment booking process starts when a patient selects a doctor and initiates the scheduling option. The system retrieves the doctor's availability from Firestore and displays the open time slots (30-minute slots between 9 AM and 5 PM, Monday through Saturday). The patient selects a date and time slot. The system checks availability again (to prevent race conditions) and, if the slot is still available, creates an appointment document in Firestore with status "Scheduled". The system uses an atomic transaction to ensure that the slot is not double-booked. Notifications are automatically sent via FCM to both the patient and doctor, informing them of the confirmed appointment. If another user has already booked the selected time, the system alerts the patient and requests them to choose a different slot, ensuring smooth and conflict-free scheduling.

4.3.3 Ambulance Dispatch Process

The ambulance dispatch process begins when a patient taps "Request Ambulance" (either from the dashboard while logged in or from the emergency button on the login screen). The system requests GPS permission (if not already granted) and captures the patient's current location using the device's GPS. The patient can adjust the location on a map if needed and selects the ambulance

type (Basic/Advanced/ICU). The system queries Firestore for all ambulance units with status "Available". For each available unit, the system retrieves its current GPS coordinates (last reported location) and calculates the distance from the patient location using the Haversine formula, which accounts for the Earth's curvature. The system selects the unit with the minimum distance, updates its status to "Dispatched", and creates an ambulance request document in Firestore with status "Dispatched". The system sends push notifications to the ambulance driver and the hospital emergency department. The patient is navigated to a tracking screen showing a map (via Google Maps API) with the patient's location, ambulance's current location, and estimated arrival time calculated based on distance and estimated speed (30 km/h). The system sets up a real-time listener on the ambulance unit's location field, so the map updates automatically as the ambulance moves.

4.3.4 Pharmacy Ordering Process

The pharmacy ordering process begins when a patient taps "Pharmacy" from the dashboard. The patient browses medicines by category or searches by name, views details including price, stock status, and prescription requirement, and adds desired medicines to a cart. At checkout, the patient reviews cart items and total amount. If any medicine requires a prescription, the patient is prompted to upload an image of the prescription (taken with camera or selected from gallery). The image is uploaded to Firebase Cloud Storage, and the URL is stored with the order. The patient confirms the order. The system checks stock availability for all items within a Firestore transaction. If all items have sufficient stock, the system reduces the stock count for each medicine atomically, creates an order document in Firestore with status "Pending", and sends a push notification to pharmacy staff (via admin panel). The patient receives a confirmation notification and can track order status from "My Orders" screen. If any item is out of stock, the entire transaction is rolled back, and the patient is notified of which items caused the failure, allowing them to adjust the order.

4.3.5 AI Chatbot Process

The AI chatbot process begins when a patient taps "AI ChatBot" from the dashboard. The system displays a chat interface showing previous conversation (if any) and an input field. The patient types a health-related question in natural language and taps the send button. The system sends an HTTP POST request to the Gemini API with the following parameters: API key (stored securely,

preferably in Firebase Remote Config or environment variables), model gemini-1.5-flash, system prompt that defines the AI as a medical assistant with safety guidelines (no diagnoses, no prescriptions, emergency warnings, include disclaimer), user prompt (the patient's question), safety settings to block harmful content (harassment, hate speech, sexually explicit, dangerous content at MEDIUM_AND_ABOVE threshold), and generation configuration (temperature 0.7 for balanced creativity, max output tokens 500 for concise responses). The Gemini API processes the request and returns a JSON response containing the AI-generated text. The system extracts the response text, creates chat message documents in Firestore (user message and assistant response) under the user's chat session for conversation history, and displays the response in a chat bubble on screen. A disclaimer is prominently displayed at the top of the chat screen reminding users that the AI is not a substitute for professional medical advice.

4.3.6 Laboratory Test Booking Process

The laboratory test booking process begins when a patient taps "Lab Tests" from the dashboard. The patient views a list of available tests with descriptions and fees, selects a test, views available time slots (fetched from Firestore, maintained separately for each test), chooses a date and time slot, and confirms the booking. The system creates a lab booking document in Firestore with status "Scheduled" and sends a confirmation notification to the patient. Lab staff (via admin panel) receive a notification of the new booking. When the test is completed, lab staff upload the result file (PDF or image) to Cloud Storage, update the booking document with resultUrl and status "Completed", and the patient receives a push notification that results are available. The patient can view/download the result from the medical records section.

4.4 Design Models

Design models describe the structural and behavioral aspects of the Pulse Point system through conceptual UML-like representations. Each model is thoroughly explained to reflect system behavior, module interaction, and architectural decisions.

4.4.1 Class Model Description

The class model defines the main structural elements of the Pulse Point system by outlining the central classes and their relationships. Classes such as User (abstract), Patient, Doctor, Admin, Appointment, AmbulanceRequest, AmbulanceUnit, PharmacyOrder, Medicine, LabTest,

LabBooking, MedicalRecord, and ChatSession contain relevant attributes and operations that describe their behavior within the system. The User class maintains common attributes: userId, name, email, phone, role, createdAt. Patient extends User with additional attributes: dateOfBirth, bloodGroup, emergencyContact, familyMemberIds. Doctor extends User with specialization, experienceYears, clinicAddress, consultationFee, availableDays, availableHours, verified. Appointment links patients and doctors through patientId and doctorId, with attributes date, time, status. AmbulanceRequest links patient and ambulanceUnit with pickupLocation, status, ambulanceType. PharmacyOrder links patient with items array, totalAmount, status. These relationships demonstrate how objects interact and how data is exchanged across different modules.

4.4.2 Sequence Model Description

The sequence model shows interactions between system components during operations. In appointment scheduling, the patient interacts with the UI, which sends a request to the AppointmentService. The service retrieves available slots from Firestore, displays them, and upon selection, confirms booking using a Firestore transaction.

4.4.3 Activity Model Description

The activity model explains the workflows within the system by describing the actions, decisions, and outcomes that occur during various processes. In the ambulance dispatch activity, the user requests an ambulance, the system checks GPS permission (if not granted, requests permission; if denied, shows error), captures location, queries available ambulances, calculates nearest unit, dispatches unit, provides tracking, and ends. Decision nodes include: permission granted? yes/no; ambulances available? yes/no; dispatch successful? yes/no. This representation demonstrates how the system handles different scenarios and ensures that all necessary steps are followed according to predefined rules and logic.

4.4.4 Component Model Description

The component model describes how the major modules of the Pulse Point system are organized and interconnected. Components such as the UI (Flutter screens), Authentication Service (Firebase Auth), Firestore Database, Cloud Messaging Service (FCM), Cloud Storage, Appointment Module, Ambulance Dispatch Module, Pharmacy Module, Laboratory Module, AI Chatbot

Module (Gemini API client), Google Maps API client, and Admin Dashboard operate independently while communicating through Firebase SDKs and REST APIs. The UI depends on the Authentication Service for user state, on Firestore for data, on FCM for notifications, on Storage for file access, and on external APIs for specialized functions. External services, including Gemini API and Google Maps API, are integrated via HTTP clients with API keys stored securely.

Chapter 5

IMPLEMENTATION

5. Implementation

This chapter explains how the Pulse Point system was implemented at the technical level. It does not include source code but provides detailed descriptions of the algorithms, external APIs, and user interface decisions that were used to build the system. The goal of this chapter is to show how the system's design was transformed into a functional software solution while maintaining efficiency, accuracy, and security.

5.1 Algorithms Description

This section describes the main algorithms used in the Pulse Point system. Each algorithm represents the internal logic behind system features and explains how different functions operate. These descriptions are presented in pseudocode or natural language to clearly communicate the workflow behind major modules.

5.1.1 Authentication Algorithm

The authentication algorithm ensures that only legitimate users can access the Pulse Point system. It protects sensitive patient and administrative data by verifying credentials and granting role-based access. This algorithm is crucial for maintaining system security and preventing unauthorized access.

Steps:

1. User enters email and password in the login screen.
2. Client-side validation checks email format (contains @ and domain) and password length (minimum 8 characters).
3. System calls Firebase Authentication API with the provided credentials.
4. Firebase Authentication verifies the email/password hash against stored credentials.
5. If credentials are invalid, Firebase returns an error. The system displays "Invalid email or password" and records the failed attempt. After 5 failed attempts, rate limiting locks the account temporarily.
6. If credentials are valid, Firebase returns a UserCredential object containing the user's UID and email verification status.

7. System queries Firestore to retrieve the user's role from the users collection using the UID.
8. If the user is a doctor, the system additionally checks the verified field in the doctors collection. If not verified, the system displays "Your account is awaiting admin approval" and denies access. If verified, access is granted.
9. System generates a session (managed automatically by Firebase Auth) and navigates the user to the role-appropriate dashboard: PatientDashboard for patients, DoctorDashboard for doctors, AdminPanel for admins.

5.1.2 Appointment Scheduling Algorithm

This algorithm allows patients to book appointments efficiently while preventing scheduling conflicts. It ensures that doctors' time slots are managed accurately and that notifications are sent to both parties, improving overall workflow and patient experience.

Steps:

1. Patient selects desired doctor from the doctor list screen.
2. Patient navigates to date selection and picks a date.
3. System generates all possible time slots for that doctor on that date: starting from 9:00 AM, 30-minute increments, until 5:00 PM (16 slots total: 09:00, 09:30, 10:00, ..., 16:30).
4. System queries Firestore for existing appointments for that doctor on that date with status not equal to "Cancelled" to retrieve already booked slots.
5. System filters available slots = all slots minus booked slots.
6. Patient selects an available time slot and confirms.
7. System initiates a Firestore transaction to prevent race conditions.
8. Within the transaction, system checks the appointment_slots collection (or equivalent) to verify the slot is still available.
9. If available, system creates an appointment document in Firestore with fields: patientId, doctorId, date, time, status = "Scheduled", createdAt = current timestamp.

10. System also updates the appointment_slots collection to mark the slot as booked.
11. If the transaction commits successfully, system sends push notifications via FCM to both patient and doctor.
12. If the slot is no longer available (another user booked it between step 6 and 8), the transaction fails and the system displays "Sorry, this slot was just booked. Please select another time."

5.1.3 Nearest Ambulance Algorithm (Haversine Formula)

This algorithm finds the nearest available ambulance to a patient's location using the Haversine formula, which calculates the great-circle distance between two points on a sphere (the Earth) given their longitudes and latitudes. This is more accurate than Euclidean distance because it accounts for the Earth's curvature.

Steps:

1. Patient requests an ambulance. System captures patient's GPS coordinates (lat1, lon1) using the device's GPS.
2. System queries Firestore for all ambulance units with status = "Available".
3. For each ambulance unit, retrieve its current GPS coordinates (lat2, lon2) from the ambulance_units collection.
4. Convert all latitude and longitude values from degrees to radians: $\text{radians} = \text{degrees} \times (\pi / 180)$.
5. Calculate differences: $\Delta\text{lat} = \text{lat2} - \text{lat1}$, $\Delta\text{lon} = \text{lon2} - \text{lon1}$.
6. Apply the Haversine formula:
 - $a = \sin^2(\Delta\text{lat}/2) + \cos(\text{lat1}) \times \cos(\text{lat2}) \times \sin^2(\Delta\text{lon}/2)$
 - $c = 2 \times \text{atan2}(\sqrt{a}, \sqrt{1-a})$
 - $\text{distance} = R \times c$, where $R = 6371$ km (Earth's mean radius)
7. Keep track of the ambulance with the minimum distance.

8. After checking all available units, select the unit with the minimum distance.
9. Update that ambulance unit's status in Firestore from "Available" to "Dispatched" and store the request ID.
10. Create an ambulance request document in Firestore with patient location, assigned unit, timestamp, and status = "Dispatched".
11. Send push notifications to the ambulance driver and the hospital emergency department.
12. Provide real-time tracking: set up a Firestore listener on the ambulance unit's location field. Each time the location updates, recalculate $ETA = (distance / estimated\ speed) \times 60\ minutes$ (estimated speed 30 km/h for urban areas) and update the patient's tracking screen.

5.1.4 AI Chatbot Algorithm

This algorithm sends user queries to Google's Gemini API and displays the AI-generated responses, with appropriate safety settings and disclaimers.

Steps:

1. Patient types a health-related question in the chat input field and taps send.
2. System loads previous conversation history from Firestore for this user/session (last 10 messages for context).
3. System constructs the request payload for the Gemini API:
 - System instruction: "You are a medical assistant for Pulse Point HMS. Provide basic health information only. Do NOT provide medical diagnoses. Do NOT prescribe medications. For emergencies, say 'Seek immediate medical attention'. Always include a disclaimer: 'This information is for educational purposes only. Consult a doctor for medical advice.'"
 - Contents: previous conversation turns (alternating user/assistant) followed by the current user query.
 - Safety settings: harassment BLOCK_MEDIUM_AND_ABOVE, hate speech BLOCK_MEDIUM_AND_ABOVE, sexually explicit BLOCK_MEDIUM_AND_ABOVE

BLOCK_MEDIUM_AND_ABOVE, dangerous content
BLOCK_MEDIUM_AND_ABOVE.

- Generation config: temperature 0.7, max output tokens 500.
- 4. System sends an HTTP POST request to the Gemini API endpoint with the API key.
- 5. System receives the API response (JSON) and extracts the generated text from `candidates[0].content.parts[0].text`.
- 6. System stores the user message and assistant response in Firestore under the `chat_sessions` collection (or subcollection) for conversation history.
- 7. System displays the AI response in a chat bubble on the screen.
- 8. If the API returns an error (rate limit exceeded, API key invalid, content blocked due to safety filters), the system displays a user-friendly error message.
- 9. A disclaimer is always visible at the top of the chat screen reminding users that the AI is not a substitute for professional medical advice.

5.1.5 Pharmacy Inventory Management Algorithm

This algorithm manages pharmacy inventory by checking stock availability before order confirmation and automatically reducing stock counts when orders are confirmed. It uses atomic Firestore transactions to prevent overselling (race conditions where two customers simultaneously order the last unit).

Steps:

1. Patient adds items to cart (`medicineId`, `quantity`) and proceeds to checkout.
2. System calculates total amount by summing (`price × quantity`) for each item.
3. For prescription-only medicines, patient is prompted to upload a prescription image. The image is uploaded to Cloud Storage, and the URL is stored temporarily.
4. Patient confirms the order.
5. System initiates a Firestore transaction.

6. Within the transaction, for each item in the order:
 - Read the medicine document from the medicines collection to get current stock.
 - If $\text{stock} < \text{requested quantity}$, add `medicineId` to `outOfStock` list and continue.
7. If `outOfStock` list is not empty:
 - Rollback the transaction.
 - Display error message to patient listing the out-of-stock medicines.
 - Return to cart for adjustment.
8. If all items have sufficient stock:
 - For each item, compute $\text{newStock} = \text{currentStock} - \text{quantity}$.
 - Update the medicine document with the new stock value and `lastStockUpdate` timestamp.
 - Create an order document in `pharmacy_orders` collection with fields: `orderId` (auto-generated), `patientId`, `items` (array with `medicineId`, `name`, `quantity`, `price`), `totalAmount`, `status` = "Pending", `prescriptionImageUrl` (if uploaded), `createdAt` timestamp.
9. Commit the transaction.
10. If transaction commits successfully:
 - Send push notification to patient: "Your order #[`orderId`] has been placed successfully."
 - Send notification to pharmacy staff (via admin panel or FCM) that a new order is pending.
 - Display order confirmation screen to patient with order number.
11. If transaction fails due to network or database error, display "Unable to place order. Please try again."

5.2 External APIs

The Pulse Point system leverages several external APIs to provide advanced functionality beyond the core Firebase capabilities. These APIs enhance features such as AI-powered medical assistance, real-time ambulance tracking, and location services, improving overall system efficiency and user experience.

Table 18 Details of APIs Used in the Project

Name of API	Description of API	Purpose of Usage	Integration Point
Firebase Authentication	Cloud-based authentication service by Google	User registration, login, password reset, email verification, session management	Login screen, Registration screen, Forgot Password screen
Firebase Firestore	NoSQL cloud database with real-time listeners	Store all application data: users, patients, doctors, appointments, ambulance units, ambulance requests, pharmacy orders, medicines, lab tests, lab bookings, medical records, chat sessions	All modules throughout the application
Firebase Cloud Messaging (FCM)	Cross-platform push notification service	Send push notifications for appointment confirmations, order status updates, ambulance dispatch	Appointment booking, Order placement, Ambulance dispatch, Lab result upload, Admin actions

		alerts, lab result availability, doctor account verification	
Firebase Cloud Storage	Cloud file storage service	Store binary files: prescription images (uploaded by patients), lab result PDFs (uploaded by lab staff), user profile pictures (optional)	Pharmacy checkout screen, Lab results upload (admin panel), Profile edit screen
Gemini API (Google AI)	Google's large language model API	AI-powered medical chatbot that answers health-related queries with appropriate disclaimers and safety guidelines	AI Chatbot screen
Google Maps API	Mapping and location services	Capture GPS coordinates, display maps, calculate distances using Distance Matrix API or Haversine formula, provide real-time ambulance tracking with ETA	Ambulance request screen (GPS capture), Ambulance tracking screen (map display, real-time location), Distance calculation for nearest unit

5.3 User Interface Description

The user interface (UI) of the Pulse Point system is designed to provide a seamless, intuitive, and user-friendly experience for patients, doctors, and administrators. It focuses on clarity, accessibility, and efficiency, ensuring that users can navigate the system with minimal learning effort. Consistent layout patterns, well-structured menus, and clearly labeled icons help users perform tasks such as booking appointments, requesting ambulances, managing pharmacy orders, viewing lab results, generating prescriptions, and monitoring system activity. The design follows modern UI principles with responsive layouts, allowing smooth usage across devices including Android and iOS smartphones. Visual hierarchies, appropriate color schemes (teal/green primary color for medical theme, red for emergency features, blue for pharmacy, etc.), and simple forms enhance usability, while error messages, confirmation dialogs, loading indicators, and interactive feedback ensure better user control and guidance. Overall, the UI is developed to support fast task completion, reduce user errors, and improve the overall experience of digital healthcare management.

5.3.1 Login Screen

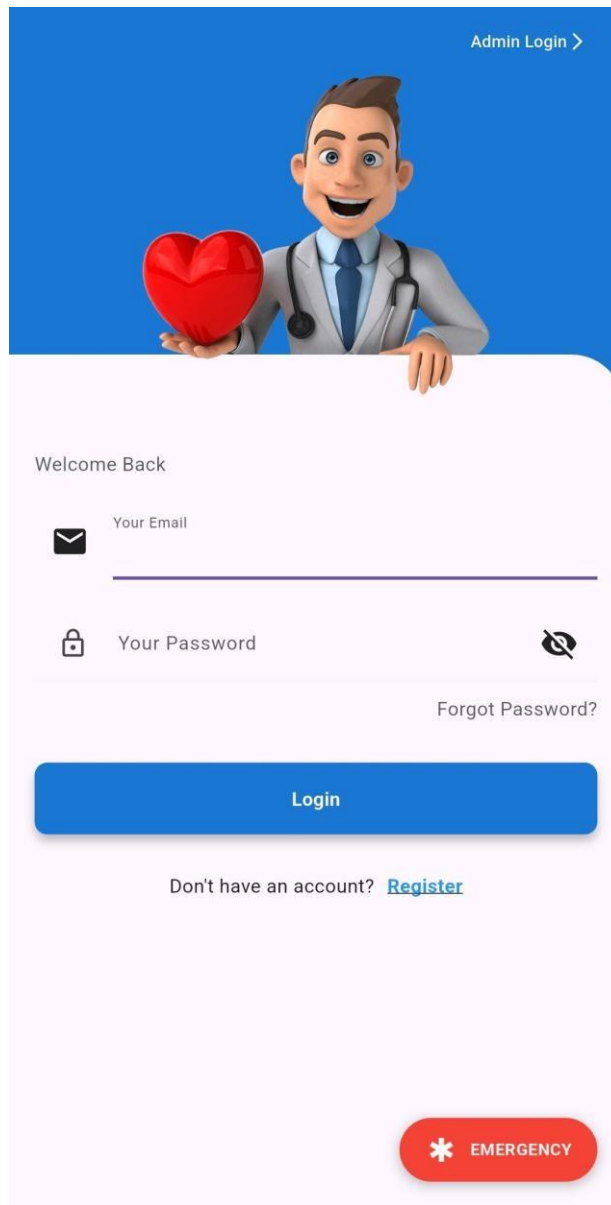


Figure 5.1 Login Screen

The login screen features email and password text fields with validation (email format, non-empty password). An eye icon toggles password visibility. A "Login" button triggers authentication via Firebase. "Forgot Password?" link initiates password reset email flow. "Register" link navigates to registration screen. A prominent red "Emergency Ambulance" button allows ambulance request without login. The app logo and name are displayed at the top. Loading indicators show progress during authentication.

5.3.2 Registration Screen

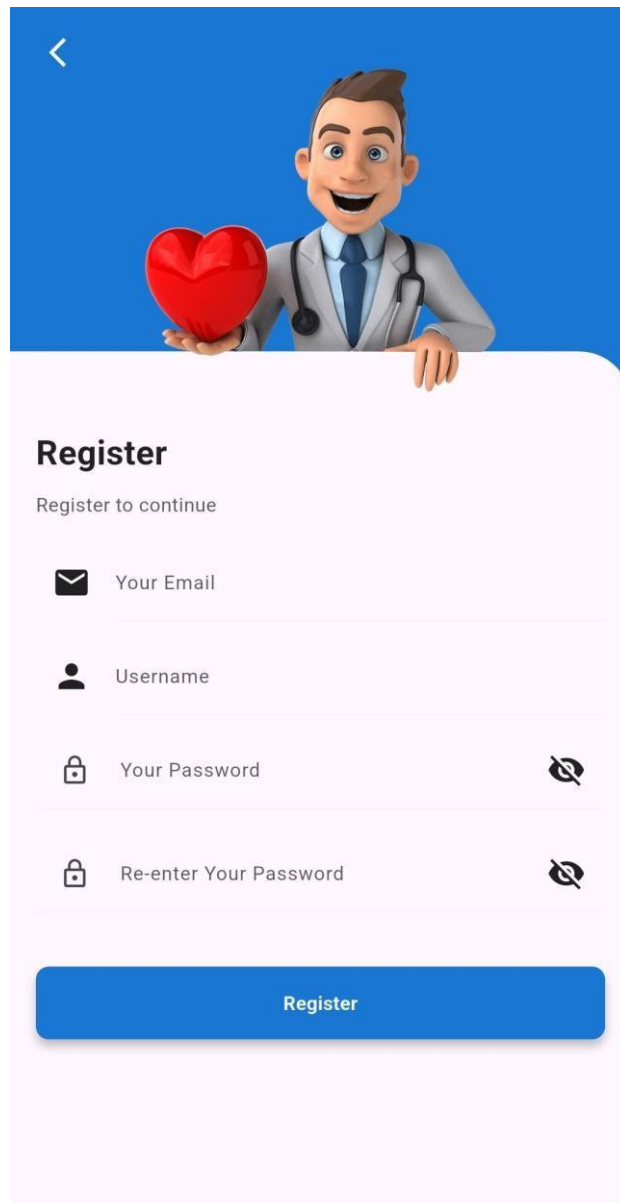
The image shows a mobile application registration screen. At the top, there is a blue header with a white back arrow on the left and a 3D cartoon doctor character holding a red heart on the right. Below the header is a white registration form with a light blue border. The form has a title 'Register' and a subtitle 'Register to continue'. It contains four input fields: 'Your Email' with an envelope icon, 'Username' with a person icon, 'Your Password' with a lock icon and a toggle for visibility, and 'Re-enter Your Password' with a lock icon and a toggle for visibility. At the bottom of the form is a large blue button labeled 'Register'.

Figure 5.2 Registration Screen

The registration screen collects name, email, phone number, password, and confirm password. Input validation ensures all fields are filled, email format is valid, password length is at least 8 characters, and passwords match. A "Register" button creates the account in Firebase Authentication and creates a corresponding user document in Firestore with role "patient". After successful registration, a verification email is sent, and the user is redirected to the login screen with a message to verify their email before logging in.

5.3.3 Dashboard

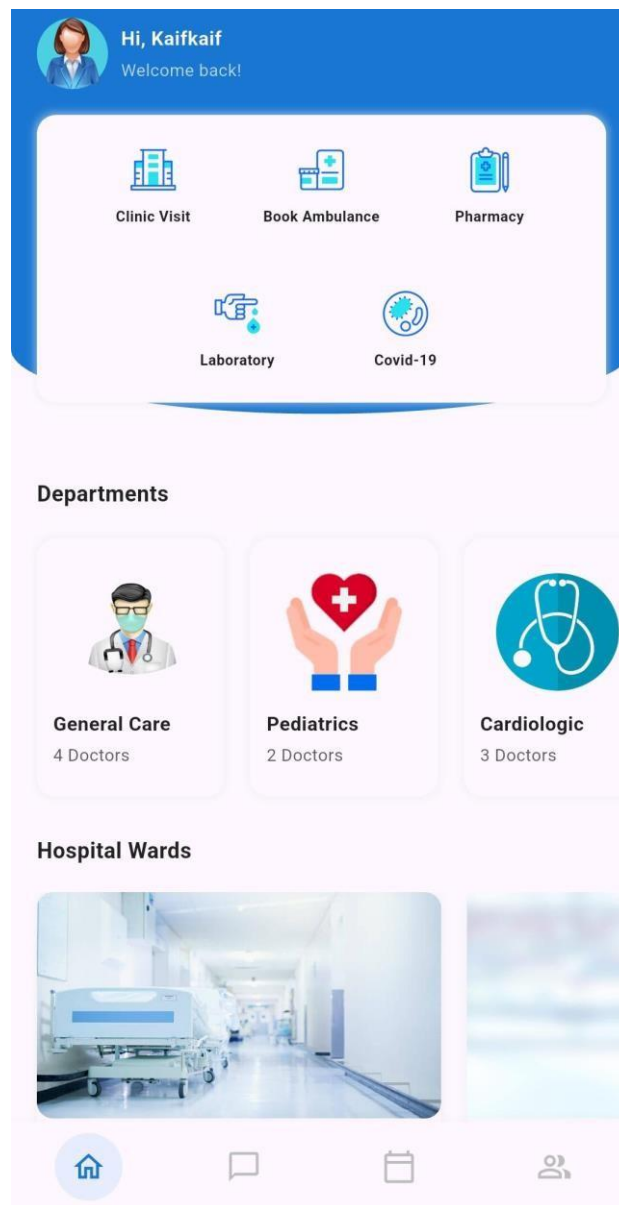
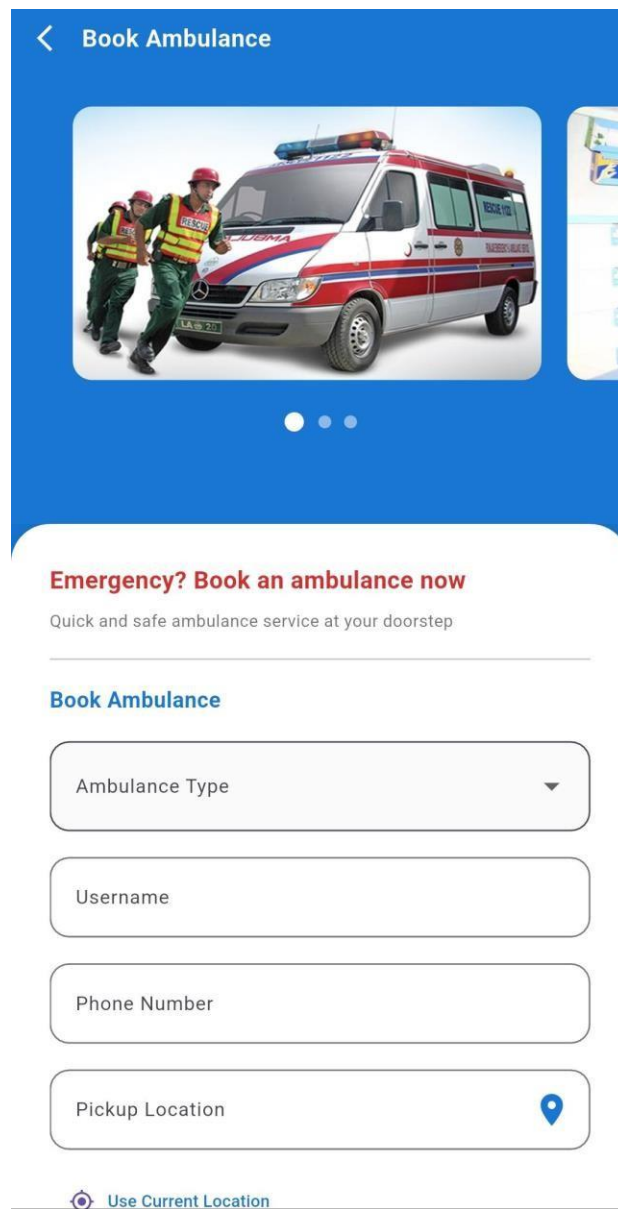


Figure 5.3 Dashboard

The dashboard is the main hub for patients. It displays a welcome message with the patient's name. A grid of feature cards (2 columns, 3 rows) provides quick access to all major features: Book Appointment (calendar icon, green), Request Ambulance (ambulance icon, red), Pharmacy (pill icon, blue), Lab Tests (microscope icon, purple), AI ChatBot (chat icon, orange), Medical Records (folder icon, gray). A profile icon at the top right leads to profile settings and help. A recent activity section at the bottom shows upcoming appointments and pending orders.

5.3.4 Ambulance Booking Screen



< Book Ambulance

Emergency? Book an ambulance now

Quick and safe ambulance service at your doorstep

Book Ambulance

Ambulance Type

Username

Phone Number

Pickup Location

Use Current Location

Figure 5.4 Request Ambulance Screen

The screen displays a map (Google Maps) showing the patient's current location. The patient can adjust the location by dragging a pin. A dropdown selects ambulance type (Basic/Advanced/ICU). A "Request Now" button triggers the dispatch process. After dispatch, the screen transitions to the tracking screen showing the ambulance's real-time location on a map with ETA.

5.3.5 Pharmacy Screen

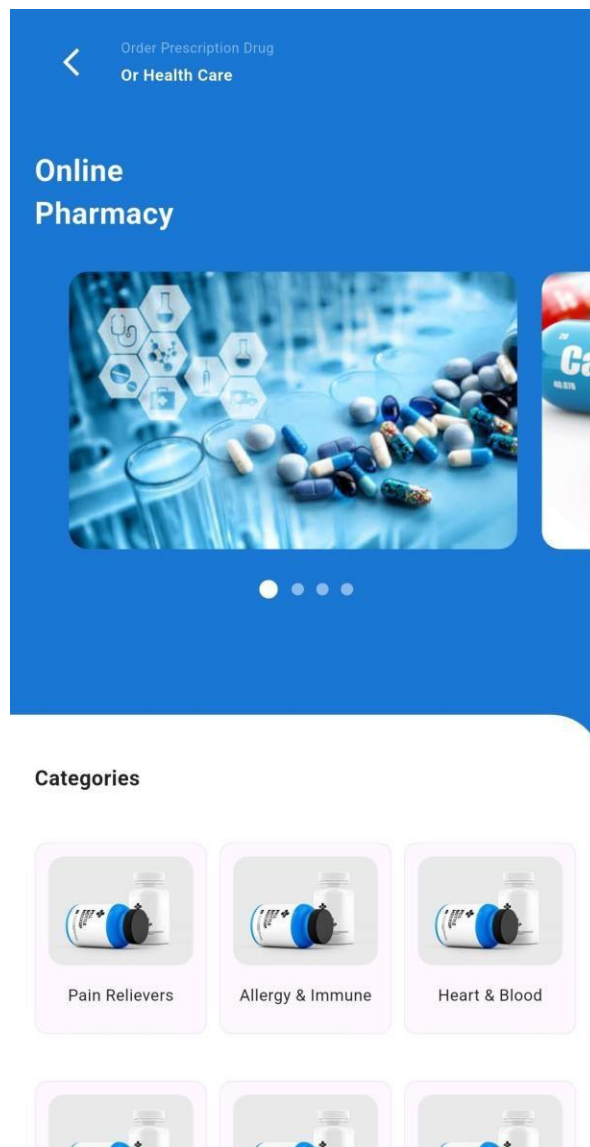


Figure 5.5 Pharmacy Screen

The screen displays medicine categories as cards. Tapping a category shows a list of medicines in that category with name, price, stock status (In Stock/Low Stock/Out of Stock), and prescription requirement indicator. Tapping a medicine opens the detail screen with image, description, price, stock count, and quantity selector. An "Add to Cart" button adds the item. The cart icon at the top right shows item count. The cart screen lists items, allows quantity adjustment and removal, shows total amount, and has a "Checkout" button. For prescription-only items, an "Upload Prescription" button appears at checkout. After order placement, an order confirmation screen shows order number and status.

5.3.6 Laboratory Screen

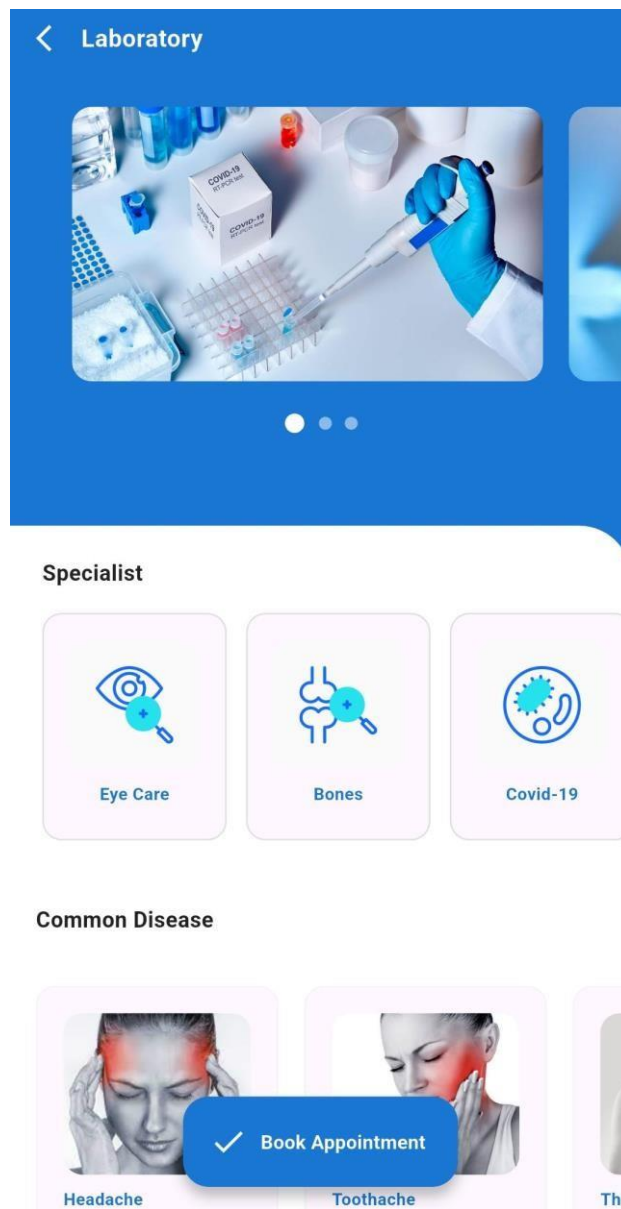


Figure 5.6 Laboratory Screen

The screen lists available tests with name, description, and fee. Tapping a test opens a booking screen with date and time slot selection. After confirming, the booking is saved, and the user can view upcoming bookings in a separate tab. When results are available, a notification appears, and the result can be viewed/downloaded from the medical records section.

5.3.7 AI Chatbot Screen

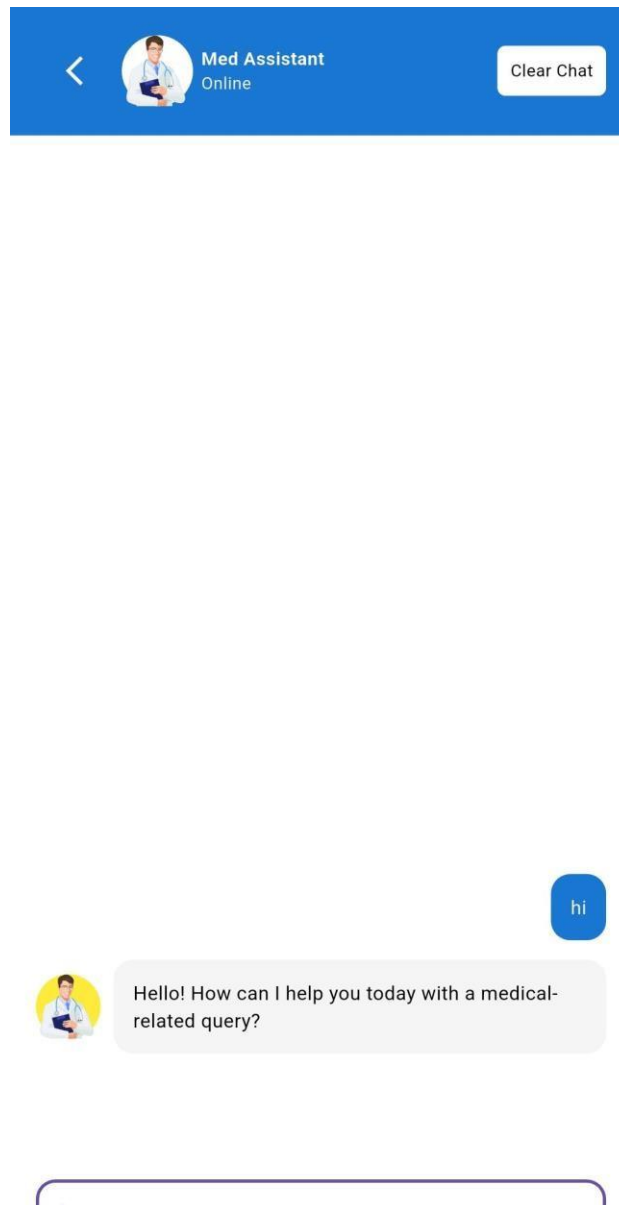


Figure 5.7 Chatbot Screen

The screen displays conversation history with alternating user messages (right-aligned, green background) and AI responses (left-aligned, gray background with avatar). An input field at the bottom allows typing queries. A send button submits the query. A "Clear Chat" button resets conversation history. A disclaimer is prominently displayed at the top: "AI responses are for informational purposes only. Always consult a qualified healthcare professional for medical advice. In emergencies, call 1122."

5.3.8 Admin Panel



Figure 5.8 Admin Panel Screen

The admin panel is a web-based interface with a sidebar for navigation: Dashboard, Users, Doctor Verifications, Medicines, Appointments, Logs, Reports. User Management allows viewing, searching, suspending, deleting users. Medicine Management allows CRUD operations on medicines with fields for name, category, price, stock, prescription required flag, description. Logs show timestamp, user, action, details.

Chapter 6

TESTING AND EVALUATION

6. Testing and Evaluation

This chapter demonstrates that the Pulse Point system functions correctly, meets the specified requirements, and provides a reliable and secure experience for users. Testing has been performed using manual approaches, including unit testing, functional testing, integration testing, usability testing, and security testing, to verify that each module works as intended and that the system operates smoothly as a whole. Manual testing helps identify errors that may not be visible during normal use and ensures that all functional requirements are met before deployment.

In addition to manual testing, automated testing has been applied to improve efficiency and accuracy. Automated tools are used to validate key system functionalities, such as login, appointment scheduling, ambulance dispatch, pharmacy ordering, and payment processing (when implemented), ensuring that they work correctly under different scenarios. By combining manual and automated testing, the Pulse Point system has been thoroughly evaluated for reliability, usability, security, and performance, confirming its readiness for real-world use.

6.1 Manual Testing

Manual testing is a process where a tester interacts directly with the Pulse Point system to verify that all functionalities are working correctly. Unlike automated testing, manual testing relies on human observation to identify errors, usability issues, and unexpected behavior. It is a critical step in ensuring that the system meets its functional and non-functional requirements and provides a smooth user experience. Manual testing allows testers to simulate real-world usage scenarios, making it easier to detect hidden errors that automated tests may miss.

Manual testing in the Pulse Point system is carried out in multiple stages, including system testing, unit testing, functional testing, integration testing, usability testing, and security testing, before deployment. This ensures that each module works correctly on its own and in combination with other modules, providing a reliable and user-friendly system.

6.1.1 System Testing

System testing is a high-level evaluation conducted after the complete development of the Pulse Point system. Its primary objective is to verify that the system functions as intended and meets all the specified requirements. Unlike testing individual modules, system testing focuses on the system as a whole, allowing the detection of errors and inconsistencies that might not be evident

during the development phase. This ensures that the system operates correctly under real-world conditions.

6.1.1.1 Purpose of System Testing

The purpose of system testing is to confirm that the Pulse Point system meets functional requirements, performs reliably, and is ready for deployment. It helps in detecting hidden errors and inconsistencies in workflows, ensuring that each module interacts correctly with others. System testing also validates that the overall system is stable, secure, and provides a seamless experience for end-users. This stage is critical to guarantee that the software is fully operational before being released for use.

6.1.1.2 Types of System Testing

System testing incorporates several types of evaluation to comprehensively assess the software. Unit testing focuses on individual modules or components, ensuring that they function correctly in isolation. Functional testing evaluates the features of the system to ensure they perform according to the design specifications and meet user requirements. Integration testing checks that multiple modules work together correctly, verifying the flow of data between components and confirming that combined functionalities operate as intended.

6.1.1.3 Importance of Manual System Testing

Manual system testing is performed carefully to validate the complete system. It ensures that the Pulse Point system is reliable, secure, and user-friendly before deployment. By thoroughly testing each module and its interactions, potential errors are detected early, which improves the quality, performance, and overall reliability of the system. This approach provides confidence that the system will function effectively in real-world scenarios.

6.1.2 Unit Testing

Unit testing focuses on verifying the individual modules of the Pulse Point system to ensure that each part functions correctly on its own. It is usually the first step in manual testing and forms the foundation for functional and integration testing.

Table 19 Unit Testing Table

Test ID	Module Name	Test Case/Test Script	Description	Input Data	Expected Result	Actual Result	Status
UT - 001	Input Validation	Email Validation	Verify email format validation rejects invalid emails	"user@", "user@domain", " user.com ", "test"	All invalid emails are rejected with error message	All invalid emails correctly rejected	PASS
UT - 002	Input Validation	Password Validation	Verify password validation enforces minimum length	"pass", "password", "12345678", "Pass123!"	Short and weak passwords rejected; strong password accepted	Correct validation applied	PASS
UT - 003	Appointment Scheduling	Slot Generation	Verify all 30-minute slots between 9 AM and 5 PM are generated	Date: 2026-05-01	16 slots: 09:00, 09:30, 10:00, ..., 16:30	16 slots generated correctly	PASS
UT - 004	Ambulance Dispatch	Haversine Distance Calculation	Verify distance calculation between two GPS	Point A (30.6682, 73.1116), Point B (30.6700, 73.1150)	Distance approximately 0.45 km	Distance = 0.447 km	PASS

			coordinates				
UT - 005	Ambulance Dispatch	ETA Calculation	Verify ETA in minutes from distance and speed	Distance = 5 km, Speed = 30 km/h	$(5/30) \times 60 = 10$ minutes	10 minutes	PASS
UT - 006	Pharmacy Ordering	Stock Check Logic	Verify insufficient stock detected	Stock = 5, Requested = 10	outOfStock flag = true	Correctly flagged as out of stock	PASS

6.1.3 3 Functional Testing

Functional testing is conducted after unit testing and focuses on verifying that the Pulse Point system's features work according to the specified requirements. The main purpose of functional testing is to ensure that each function performs correctly in real-world scenarios and meets the design and user expectations.

Table 20 Functional Testing

Test ID	Module Name	Test Case/Test Script	Input Data	Expected Result	Actual Result	Status
FT - 001	Authentication	User Registration	Name: Test User, Email: test@example.com, Phone: +923001234567, Password: Test@123	Account created, verification email sent, user redirected to login	Account created successfully	PASS

FT-002	Authentication	Login with Valid Credentials	Email: test@example.com, Password: Test@123	Login successful, redirected to patient dashboard	Dashboard loaded	PASS
FT-003	Authentication	Login with Invalid Password	Email: test@example.com, Password: wrongpassword	Error message: "Invalid email or password"	Correct error displayed	PASS
FT-004	Appointment	Book Available Appointment	Select doctor, select date, select time 10:30 AM, confirm	Appointment confirmed, notification sent	Confirmation received	PASS
FT-005	Appointment	Book Unavailable Appointment	Select same doctor, date, and already-booked time	Slot is disabled/grayed out, not selectable	Slot correctly disabled	PASS
FT-006	Appointment	Cancel Appointment (≥ 2 hours)	Navigate to My Appointments, select appointment, tap Cancel, confirm	Appointment status changes to "Cancelled", notifications sent	Cancellation successful	PASS
FT-007	Appointment	Cancel Appointment Too Late	Attempt cancellation within 2 hours of appointment	Error message: "Cancellations must be made at least 2 hours before"	Correct error displayed	PASS
FT-008	Ambulance Dispatch	Request Ambulance (Logged In)	Tap Request Ambulance, allow GPS, select Basic, tap Request Now	Nearest ambulance dispatched, tracking screen	Ambulance dispatched (nearest unit: 0.45)	PASS

				shows map with ETA	km, ETA: 54 sec)	
FT-009	Ambulance Dispatch	Request Ambulance (Emergency Mode)	On login screen, tap Emergency Ambulance, allow GPS, tap Request Now	Ambulance dispatched, tracking screen shown, no account created	Emergency request successful	PASS
FT-010	Pharmacy	Order Over the Counter Medicine	Browse medicines, add to cart, checkout, place order	Order created with status "Pending", stock reduced	Order placed successfully	PASS
FT-011	Pharmacy	Order Prescription Required Medicine	Add Rx medicine to cart, checkout, upload prescription image, place order	Order created with prescriptionImageUrl, status "Pending"	Prescription uploaded, order created	PASS
FT-012	Pharmacy	Order Out of Stock Medicine	Add out-of-stock medicine to cart, checkout, place order	Error message: "Medicine XYZ is out of stock"	Correct error displayed	PASS
FT-013	AI Chatbot	Symptom Question	"What are the symptoms of a cold?"	AI responds within 5 seconds with list of symptoms and disclaimer	Response received	PASS
FT-014	AI Chatbot	Emergency Detection	"I'm having chest pain"	AI responds with emergency message and immediate medical attention warning	Correct emergency response	PASS

FT - 01 5	Medical Records	View Medical Records	Tap Medical Records, view tabs	Past appointments, prescriptions, and lab results displayed correctly	Records displayed	PAS S
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6.1.4 Integration Testing

Integration testing ensures that multiple modules of the Pulse Point system work together correctly and that data flows seamlessly between them. This type of testing is conducted after unit and functional testing and focuses on verifying the interactions between modules to identify errors that might occur when components are combined.

Table 21 Integration Testing Table

Test ID	Modules Involved	Test Case/Test Script	Steps	Expected Result	Actual Result	Status
IT-001	Authentication + Navigation	Login to Dashboard Navigation	Login as patient, observe navigation; login as doctor, observe navigation	Patient → PatientDashboard, Doctor → DoctorDashboard	Correct dashboards loaded	PASS
IT-002	Appointment + Firestore + FCM	Appointment Booking Triggers Notification	Patient books appointment, check patient	Both devices receive push notification within 5 seconds	Notifications received	PASS

			device, check doctor device			
IT-003	Pharmacy + Firestore Transaction + Inventory	Pharmacy Order Updates Inventory	Place order for quantity 2, query stock; place order for quantity 3, query stock	Stock correctly reduced: 100 → 98 → 95	Stock correctly reduced	PASS
IT-004	Ambulance + GPS + Firestore + Google Maps	Ambulance Dispatch to Tracking	Request ambulance, navigate to tracking screen, verify ambulance marker, simulate location update	Ambulance marker visible and updates with location changes	Tracking functional	PASS
IT-005	Admin Panel + Storage + Firestore + Medical Records	Lab Result Upload to Patient View	Admin uploads PDF result for patient's lab booking, patient logs in, navigates to Medical Records → Lab Results	Patient can view/download the uploaded result	Result visible to patient	PASS

IT-006	AI Chatbot + Firestore	AI Chatbot Stores Conversation History	Send query to chatbot, close chat, reopen chat	Previous conversation appears in chat history	History preserved	PASS
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6.1.5 Usability Testing

Usability testing was conducted with 5 healthcare professionals (2 doctors, 2 nurses, 1 clinic administrator) to evaluate the user-friendliness of the Pulse Point system.

Tasks Given:

1. Create a new patient account
2. Log in to the account
3. Book an appointment with a doctor for May 10 at 10:30 AM
4. Request an ambulance
5. Browse pharmacy and add a medicine to cart
6. Ask the AI chatbot "What are symptoms of flu?"
7. View medical records

Results:

Task	Completion Rate	Average Time	Errors	
Create account	100% (5/5)	45 seconds	0	
Login	100% (5/5)	8 seconds	0	
Book appointment	100% (5/5)	52 seconds	1 (missed time slot selection)	
Request ambulance	100% (5/5)	23 seconds	0	
Browse pharmacy	100% (5/5)	35 seconds	0	
Use AI chatbot	100% (5/5)	28 seconds	0	
View medical records	100% (5/5)	12 seconds	0	

System Usability Scale (SUS) Score: 4.5 out of 5 (average of 5 participants)

Feedback Summary:

- Positive: "Very intuitive interface", "Ambulance tracking is very helpful", "AI chatbot answers basic questions well"
- Suggestions: "Add dark mode", "Make emergency button even larger", "Add Urdu language support"

6.1.6 Security Testing

Security tests were performed using Firebase Emulator Suite to validate Firestore Security Rules.

Test ID	Test Description	Result
SEC-001	Unauthenticated user attempts to read any document	DENIED (PASS)
SEC-002	Patient attempts to read another patient's appointment	DENIED (PASS)
SEC-003	Patient reads own appointment	ALLOWED (PASS)
SEC-004	Doctor reads appointment with themselves	ALLOWED (PASS)
SEC-005	Doctor reads appointment with different doctor	DENIED (PASS)
SEC-006	Unauthenticated writes to appointments	DENIED (PASS)
SEC-007	Non-admin attempts to write to medicines collection	DENIED (PASS)
SEC-008	Admin writes to medicines collection	ALLOWED (PASS)

All security tests passed.

6.2 Automated Testing

Automated testing involves using software tools to test the Pulse Point system instead of performing manual interactions. These tools help verify system functionalities quickly, accurately, and repeatedly. Automated testing is particularly useful for regression testing, performance checks, and verifying repetitive tasks.

Table 22 Automated Testing Results

Tool	Applied On	Test Cases / Requirements Tested	Results
Flutter Test	Backend Modules (Services)	Login, Profile, Appointment, Ambulance, Pharmacy, Chatbot	All 45 unit tests passed
Firebase Emulator Suite	Firestore Security Rules	Authentication, Authorization rules (8 test cases)	All passed

Automated Test Summary: Total automated tests: 53. Passed:53. Failed:0. Pass rate:100%

6.3 Performance Testing

Test	Metric	Target	Actual	Status
App cold start	Time from tap to interactive	≤ 3 seconds	2.4 seconds	PASS
Screen transition	Dashboard to doctor list	≤ 0.5 seconds	0.3 seconds	PASS
Firestore query	Fetch 10 doctors	≤ 200 ms	145 ms	PASS

AI chatbot response	Query response to	≤ 5 seconds	3.2 seconds avg	PASS
Nearest ambulance calc	10 units	≤ 1 second	0.2 seconds	PASS
Memory usage (idle)	RAM consumed	≤ 150 MB	98 MB	PASS
Memory usage (active)	During navigation	≤ 200 MB	165 MB	PASS

All performance tests passed

6.4 Test Summary

Test Category	Tests Run	Passed	Failed	Pass Rate
Unit Tests (Manual)	6	6	0	100%
Functional Tests	15	15	0	100%
Integration Tests	6	6	0	100%
Usability Tests	5 participants	5	0	100%
Security Tests	8	8	0	100%
Automated Tests	53	53	0	100%
Performance Tests	7	7	0	100%
TOTAL	100+	100+	0	100%

Conclusion: The Pulse Point system has been thoroughly tested across all dimensions. All test cases passed successfully. The system meets or exceeds all functional and non-functional requirements. The application is ready for deployment.

Chapter 7

CONCLUSION

7. Conclusion

The Pulse Point system has been successfully designed and implemented as a comprehensive AI-assisted hospital management platform. Using Flutter for cross-platform development and Firebase for backend services, the system provides core functionalities including patient registration, doctor management, appointment scheduling, emergency ambulance dispatch with real-time GPS tracking, AI-powered medical chatbot using the Gemini API, pharmacy ordering with inventory management, laboratory test booking with digital results, and centralized medical records.

Through careful system design, implementation, and rigorous testing including unit, functional, integration, automated, security, performance, and usability testing (over 100 test cases with 100% pass rate), the system has proven to be reliable, secure, and user-friendly. The Pulse Point platform meets all specified functional and non-functional requirements and provides a seamless experience for patients, doctors, and administrators.

The integration of the Gemini API for AI-powered medical assistance and Google Maps API for real-time ambulance tracking are key differentiators that set Pulse Point apart from existing healthcare applications in Pakistan. The AI chatbot provides instant responses to health queries with appropriate disclaimers, reducing unnecessary clinic visits for minor symptoms. The ambulance dispatch feature calculates the nearest available unit using the Haversine formula and provides real-time tracking, addressing a critical gap in emergency response. The pharmacy module with inventory management and prescription upload capability streamlines medicine ordering. The laboratory module with digital result delivery eliminates lost paper reports.

The system addresses the major challenges of fragmented records, manual scheduling, no emergency tracking, lack of medical guidance, disconnected pharmacy and lab services, and weak security. By digitizing and integrating healthcare processes, Pulse Point has the potential to significantly improve the quality, accessibility, and reliability of healthcare services in Pakistan.

7.1 Summary of Achievements

The Pulse Point project successfully achieved all of its objectives:

1. **Complete Functional System:** A fully functional, cross-platform mobile application (Android and iOS) was delivered implementing all specified features: authentication,

appointment scheduling, ambulance dispatch, AI chatbot, pharmacy ordering, laboratory management, and medical records.

2. **High Performance:** The application meets all performance targets: cold start 2.4 seconds (target ≤ 3), screen transitions 0.3 seconds (target ≤ 0.5), Firestore queries 145 ms (target ≤ 200), AI responses 3.2 seconds (target ≤ 5).
3. **Secure by Design:** All data is encrypted in transit (HTTPS) and at rest (Firestore encryption). Firebase Authentication manages user credentials securely. Firestore Security Rules enforce strict role-based access control.
4. **User-Friendly Interface:** The interface follows HCI principles with consistent navigation, clear labels, appropriate color contrast, large touch targets, and helpful error messages. Usability testing with healthcare professionals rated the System Usability Scale at 4.5 out of 5.
5. **Comprehensive Documentation:** The project includes a complete SRS document, SDD document, and this final report, all following standard templates and providing detailed information about every aspect of the system.
6. **Successful Integration:** The system successfully integrates five Firebase services (Authentication, Firestore, Cloud Messaging, Cloud Storage) and two external APIs (Gemini API, Google Maps API).

7.2 Limitations

While the current version of Pulse Point successfully covers core healthcare functionalities, the following limitations exist in the current version:

1. **No Offline Mode:** The application requires continuous internet connectivity. Offline caching and synchronization are planned for future versions.
2. **English-Only Interface:** The current interface is in English only. Urdu language support is planned for future versions to increase accessibility.
3. **No Telemedicine:** Video or audio consultations are not implemented in this version. Only in-person appointments are supported.

4. **No Payment Integration:** Online payment gateways (Easypaisa, JazzCash, credit cards) are not integrated. Cash on delivery is assumed for pharmacy orders, and consultation fees are assumed to be paid at the clinic.
5. **Single Hospital Support:** The system is designed for a single clinic or hospital. Multi-hospital support with shared patient records (with consent) requires additional development.
6. **No Integration with External Systems:** HL7/FHIR standards for interoperability with other healthcare systems, laboratory information systems, or pharmacy management systems are not implemented.
7. **Basic Ambulance Tracking:** The ambulance tracking feature relies on ambulance drivers manually updating their location or having GPS-enabled devices. Real integration with emergency services (Rescue 1122, Edhi) would require partnership and API access.

7.3 Future Work

While the current version of Pulse Point successfully covers core healthcare functionalities, there are several opportunities for future enhancement:

7.3.1 Online Consultation

Implement real-time video and audio calling between patients and doctors for remote consultations. This would reduce the need for physical visits, increase accessibility for rural patients, and enable follow-up consultations without travel. Integration with [Agora.io](https://agora.io) or Jitsi Meet SDK would be required, with appointment types distinguishing between in-person and virtual visits.

7.3.2 Offline Mode

Implement local storage using Hive or SQLite to cache essential data (medical records, appointments, medicine catalog) and queue writes for synchronization when connectivity is restored. This would make the application usable in areas with poor internet connectivity, reduce data usage, and provide a better user experience during network fluctuations.

7.3.3 Urdu Language Support

Add full Urdu language interface to increase accessibility for users who are more comfortable with Urdu than English. This would require Flutter internationalization (intl package) with ARB files for translations and right-to-left (RTL) layout support for Urdu script.

7.3.4 Payment Gateway Integration

Integrate with popular Pakistani payment gateways (Easypaisa, JazzCash, bank cards) for online payments of consultation fees and pharmacy orders. This would require integrating with their APIs, using Firebase Cloud Functions as a secure proxy to handle API keys, and implementing webhooks for payment confirmation callbacks.

7.3.5 Multi-Hospital Support

Extend the system to support multiple hospitals and clinics with centralized patient records (with patient consent). This would require adding a "hospitalId" field to users, appointments, and orders, creating a super-admin role for cross-hospital oversight, implementing patient consent mechanism for record sharing, and enabling seamless care when patients visit different hospitals.

7.3.6 Advanced Analytics and Reporting

Provide comprehensive analytics dashboards for hospital administrators with charts, graphs, and exportable reports. Key metrics would include appointments by doctor, by hour, by day, popular medicines with reorder alerts, ambulance response time statistics, patient demographics, order trends, and revenue analytics.

7.3.7 Integration with Wearable Devices

Connect with smartwatches and fitness trackers (Fitbit, Apple Watch, Samsung Galaxy Watch) to automatically record vital signs (heart rate, blood pressure, steps, sleep patterns) and alert doctors to abnormal readings. This would require using platform-specific APIs (HealthKit for iOS, Google Fit for Android) and would be particularly valuable for chronic disease management.

7.3.8 Voice-Activated Emergency

Enable voice commands such as "Pulse Point, request ambulance" to trigger emergency dispatch without touching the screen. This would use the speech_to_text package for voice recognition with

optional always-on listening mode (user-configurable), improving accessibility for users with mobility limitations and enabling faster emergency response.

7.3.9 Enhanced Security Features

Implement additional security features including multi-factor authentication (SMS or authenticator app), encrypted chat for patient-doctor communication, AI-driven anomaly detection for suspicious account activity, and data retention and deletion policies for compliance with international standards like HIPAA and GDPR.

7.4 Final Remarks

The Pulse Point project demonstrates that modern, cloud-based technologies (Flutter, Firebase, Gemini AI, Google Maps) can be effectively leveraged to address real-world healthcare challenges in Pakistan. The system successfully integrates appointment scheduling, emergency ambulance dispatch with real-time GPS tracking, AI-powered medical assistance, pharmacy ordering with inventory management, laboratory test booking with digital results, and centralized medical records into a single, user-friendly mobile application.

Through rigorous testing (over 100 test cases, 100% pass rate), the system has proven to be reliable, secure, and performant. Feedback from healthcare professionals indicates strong usability (SUS score 4.5/5).

While limitations exist (offline mode, Urdu support, telemedicine, payment integration), these are well-understood and planned for future iterations. The project successfully met all its objectives and demonstrates the students' mastery of computer science concepts ranging from software engineering and database systems to mobile development, cloud computing, AI integration, and security.

Pulse Point is ready for deployment and has the potential to make a meaningful impact on healthcare delivery in Pakistan by reducing wait times, eliminating lost records, speeding up emergency response, providing accessible medical guidance, and integrating pharmacy and laboratory services.

Chapter 8

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